

Introduction to Applied Data Science Course

Review and Introduction to Applied Data Science

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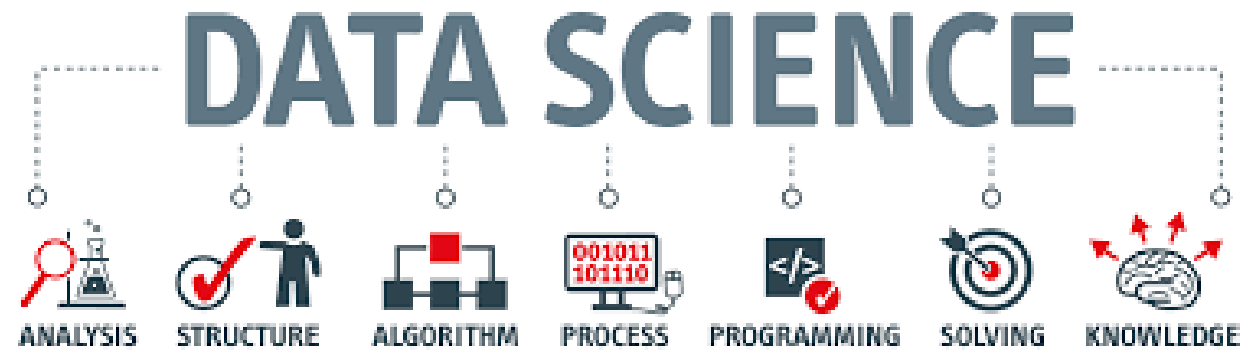
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- ◎ **What is Data Science?**
- ◎ About course
- ◎ Review
 - Collecting data
 - Understanding data
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 - Machine learning
 - IT Infrastructure

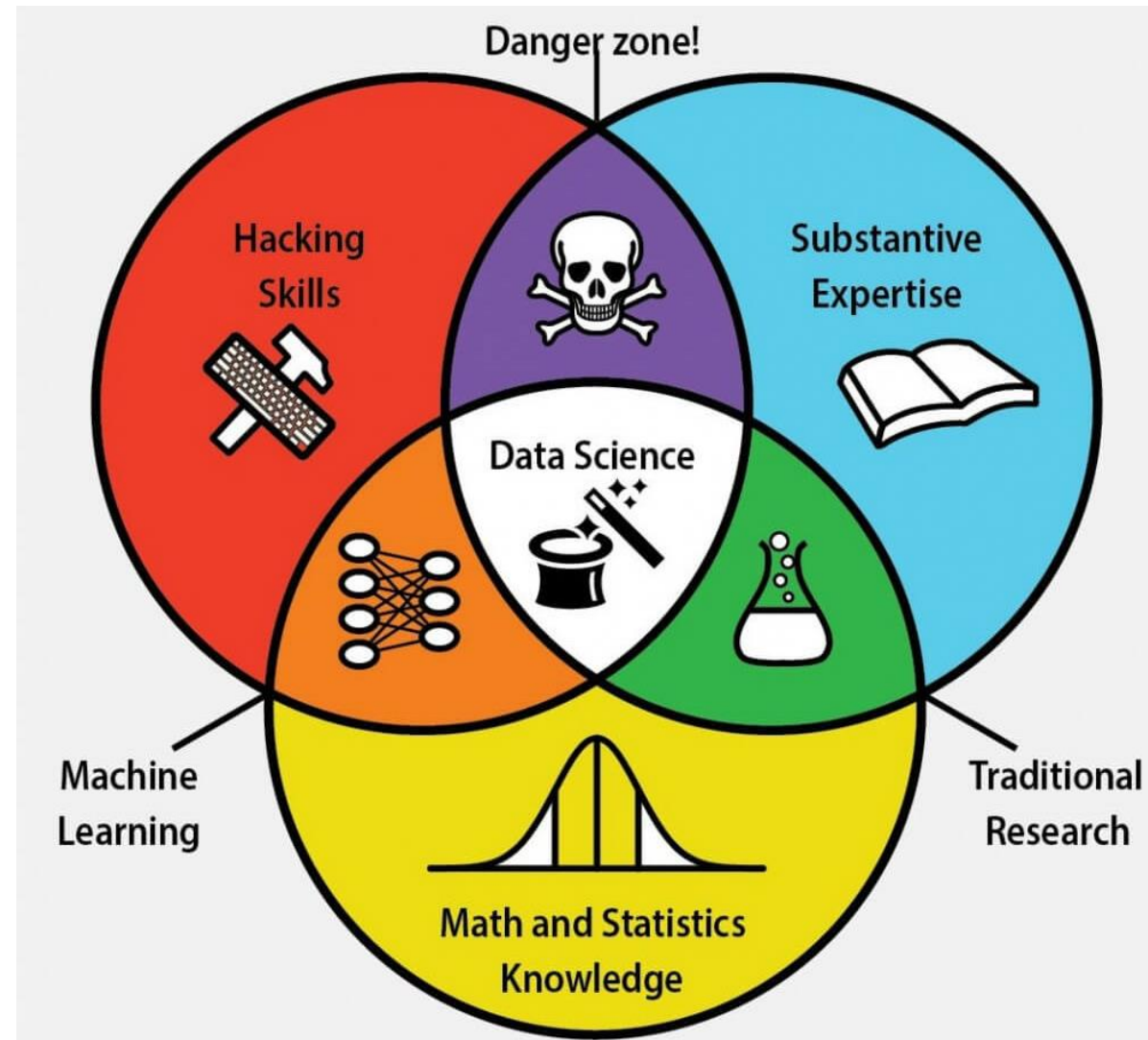
What is Data Science?

◎ **Extracting useful conclusion** from large datasets with:

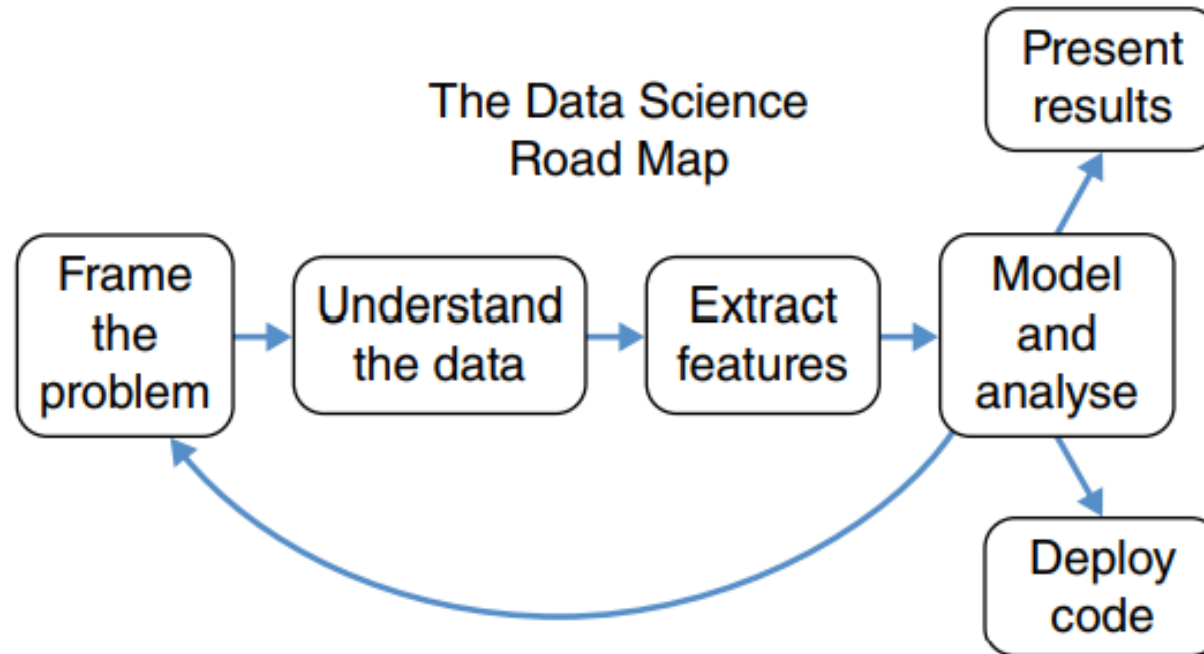
- Processes
- Problem definition
- Set of principles
- Analysis
- Algorithms
- Computer



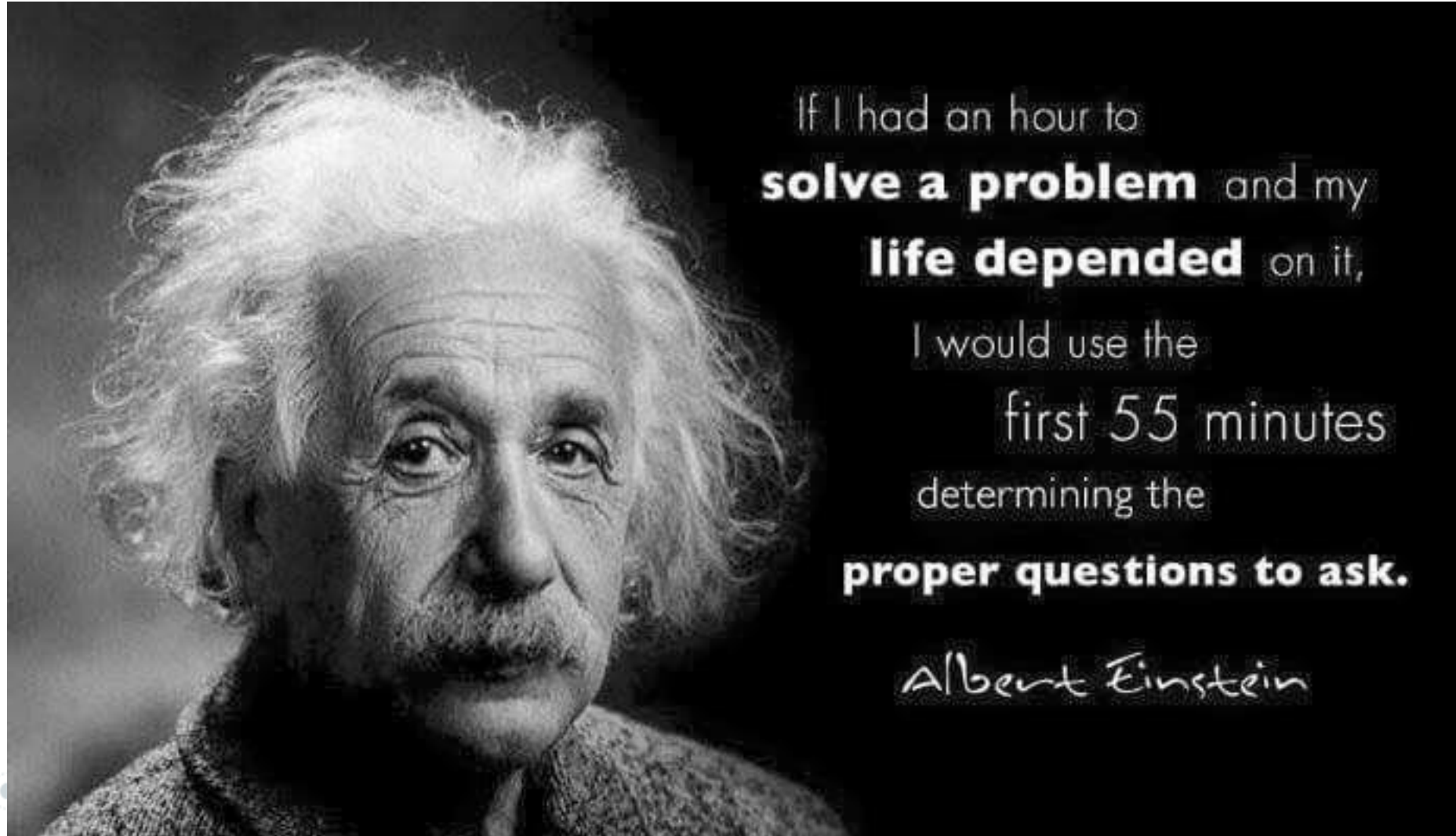
Skills-set for a data scientist



What is DS-ist do?



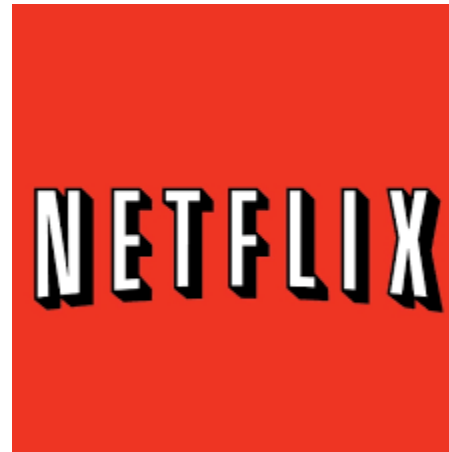
Data Scientist ~ “Frame is problem”



Where Is Data Science Used?

◎ Case studies that illustrate the impact of data science:

- **Consumer companies:** sales and marketing
 - Walmart
 - Netflix
 - Amazon

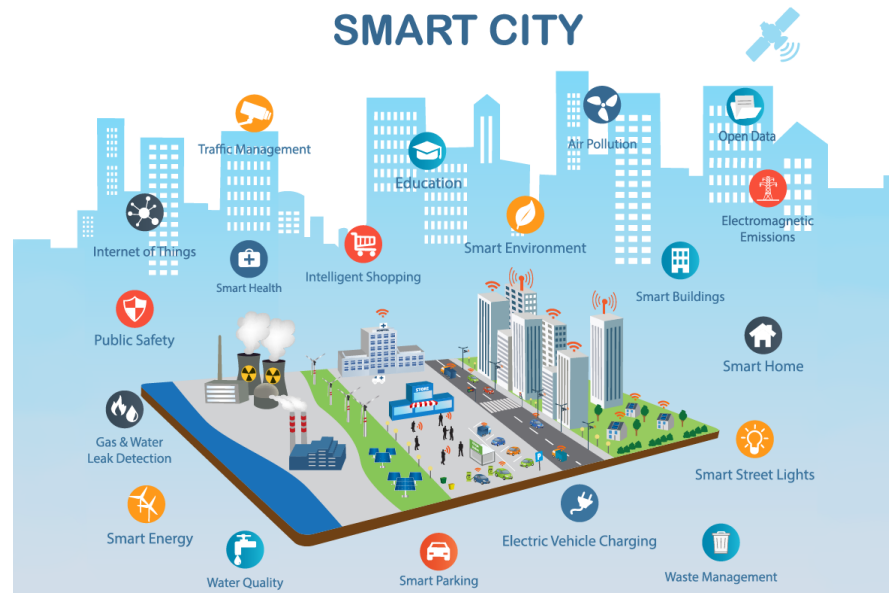


Where Is Data Science Used?

◎ Case studies that illustrate the impact of data science:

○ Governments

- Organize cities: track, analyze, and control environmental, energy, and transport systems and to inform long-term urban planning
- Design drugs for individual patients
- Predict crime hot spots and recidivism



Where Is Data Science Used?

◎ Case studies that illustrate the impact of data science:

- Consumer companies: sales and marketing
- Governments: improve health, criminal justice, and urban planning
- Professional sporting:
 - ◎ Strategies



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Course Assessment

◎ Contacts:

- Le Ngoc Thanh and tutors
- Communication channel: Moodle, Email, Social Network

◎ Course Assessment:

- Exercises/Classwork/Homework: 30%
- Mini projects: 20%
- A competition project: 50%

◎ Regularizations:

- Attend classes regularly and have discussions
- If total score of exercises/classwork/homework/mini projects below 2.5 points (base 10), the competition project will not count.
Any plagiarism will be 0 points for course grade.

Team

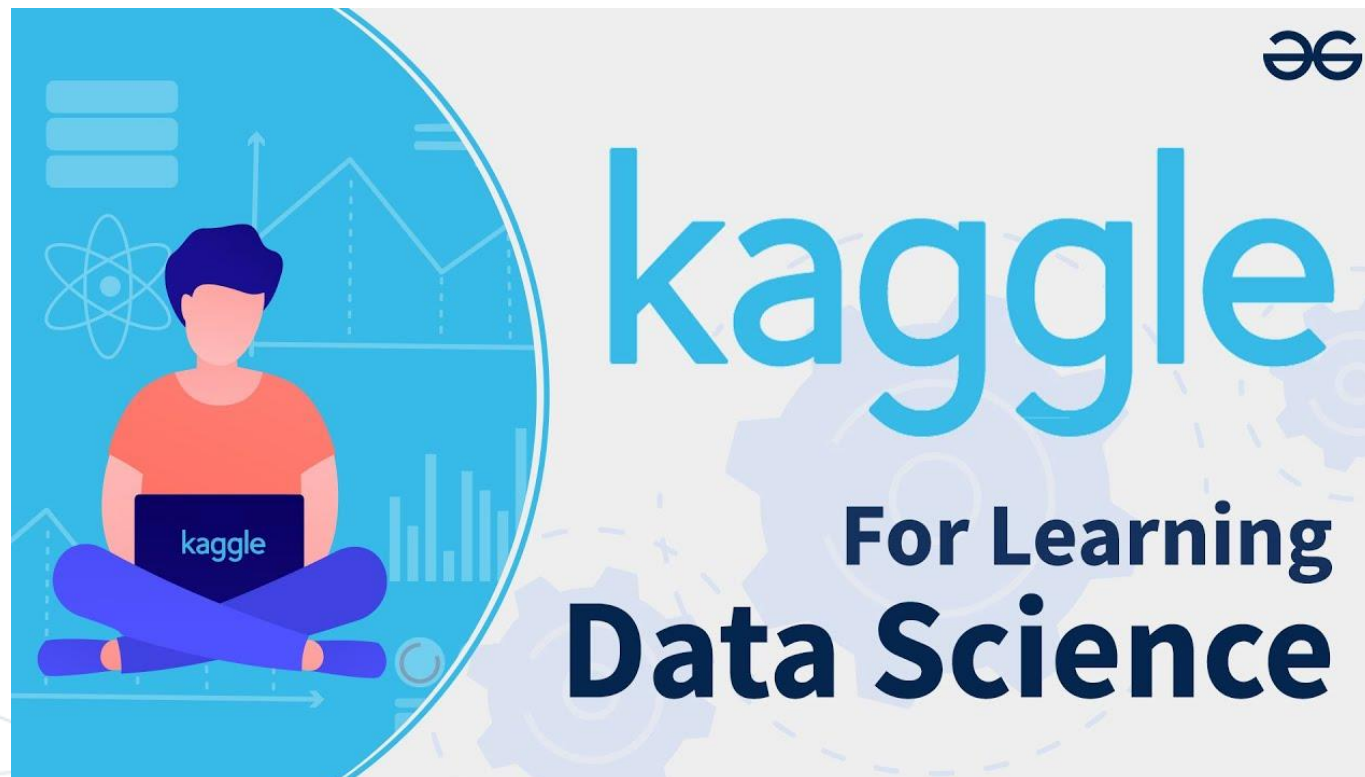
- ◎ Each team includes **maximum of 4 students.**
- ◎ Preparation:
 - Build a team by yourself
 - Choose a team leader
 - Create a team name
 - Make a plan

Mini-Projects

- ◎ This is teamwork.
- ◎ Each week (with the exception of midterm and final presentations), at least one member presents about required topics on Moodle.
- ◎ Prepare responses to the provided questions.
- ◎ Ensure that the content does not duplicate that of other groups.
- ◎ Utilize Google Colab to demonstrate the procedure in detail. The team must manually type and explain each step, not click run on the existing code. Students can view materials during the demonstration.
- ◎ Share your notebooks with QR-Code so that everyone in the class can provide feedback.
- ◎ After the presentation, the group recount the number of comments for each student.

Topics on Kaggle

- ◎ The majority of weekly topics are available on Kaggle.
- ◎ The student may write the source code by yourself or utilize the available source code; nonetheless, you must comprehend each stage of implementation.



The Most Comprehensive List of Kaggle Solutions and Ideas

This is a list of almost all available solutions and ideas shared by top performers in the past Kaggle competitions. This list gets updated as soon as a new competition finishes. If you find a solution besides the ones listed here, I would encourage you to contribute to this repo by making a pull request. The symbols used in this list are described [here](#).

If you found it interesting, you may give a star or make a fork



2,514



812

Check out the following markdown pages about Top Kagglers Tips/Tricks and all Kernels of The week.

- [Top Kagglers Interviews and Lectures](#)
- [Kernels of The Week](#)

Last Updated: February 1, 2023

	Title & Description	Details	Solutions	Pins
	<u>523. Lux AI 2022 - Beta</u> Terraform mars and help test season 2 of the Lux AI Challenge!	Prize: Swag Team: 44 Kind: Playground Metric: Year: 2022		
	<u>522. DFL - Bundesliga Data Shootout</u> Identify plays based upon video footage	Prize: \$25,000 Team: 530 Kind: Featured	1st place 2nd place 3rd place	

The Competition Project

- ◎ Each group chooses a competition topic (data in table form) to work on.
- ◎ The topic is certain to have no public code and is currently in development (ongoing).
- ◎ The student may select topics that have been completed within the past two to three months and must ensure that the source code has not been published.
- ◎ Your topic does not overlap with another group.
- ◎ The team registers for competitions and demonstrates your efforts.
 - If the score falls between 10 and 30, bonus points will be awarded.
 - If your project ranks among the top 10 in the current competition's public mode, you will receive full project points after finishing presentations.

Midterm and Final term presentations

© Midterm tasks:

- Describe the problem's characteristics and challenges.
- Present about data and its attributes.
- Conduct data preprocessing, data visualization, and statistical analysis.
- A summary of related approaches, your justifications, and a pros and cons analysis of each approach.
- Determine and explain the primary direction to do.
- Each member must comprehend the work of the others and actively participate in the presentation. The instructor will ask any student about any content. Other members can add comments.

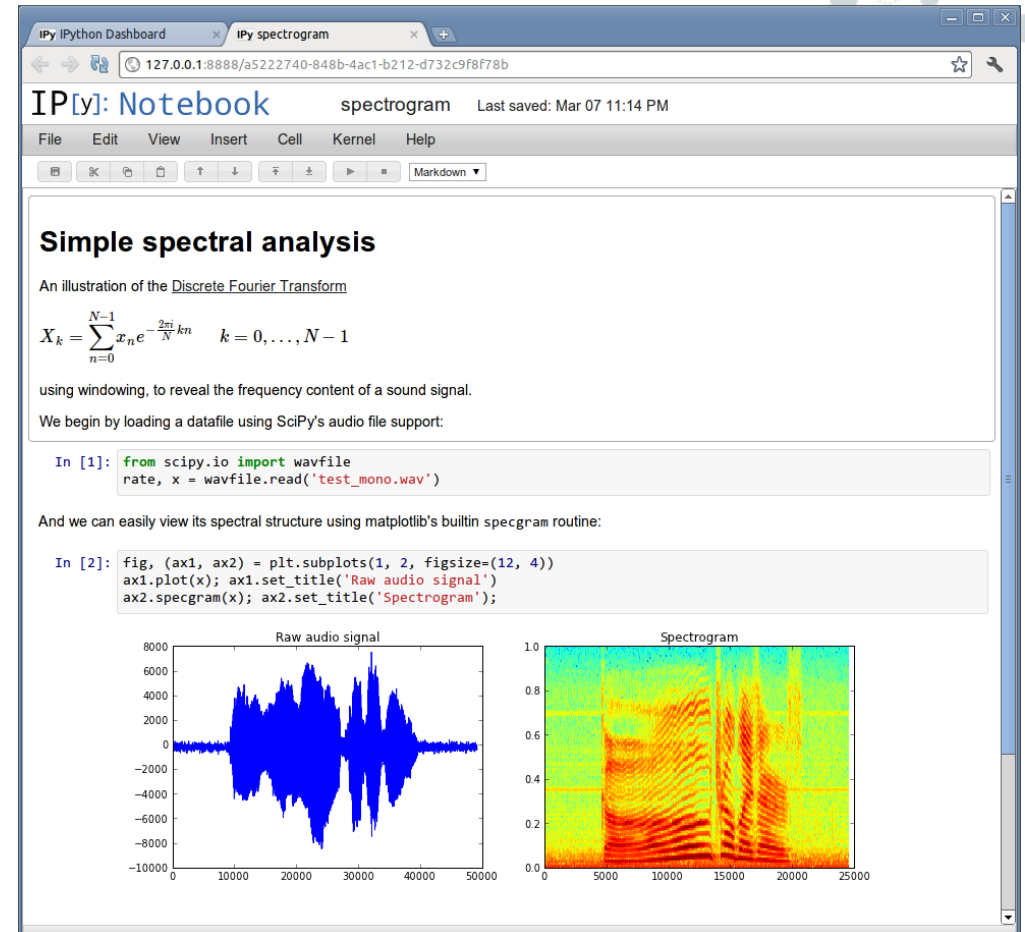
Midterm and Final term presentations

© Final term tasks:

- The team must specify the solution method.
- It is recommended to compare two selected implementation methods.
- Describe the measure and the experiment and evaluation procedures.
- Display and illustrate the results.
- Review and evaluate the achieved results.
- Evidence of ranking on Kaggle.
- All members must present.

Tools

- ◎ Python and the Libraries
- ◎ Google Colab
- ◎ Jupyter Notebook



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Collecting data

◎ General notes when collecting data

- Is the data correct and sufficient to answer the question?
Garbage in → garbage out
- Is collecting such data valid? Does it affect others?

◎ Ways to collect Data

- Data is **available** in **company, organization**: ok, use it
- Data is **available** but **out there** (online)
 - Pre-packaged data (file csv, excel, ...): download
 - Data provided through the website's API: use API
 - Data is on the site but no API: parse HTML

The data is **not yet available**: created by yourself in ways such as conducting surveys, using sensor devices, ...

Scope
of the
course

Use basic requests-HTML libraries (document lookups as needed)

◎ Import the library

```
from requests_html import HTMLSession
```

◎ Get the website's HTML code

```
session = HTMLSession()
```

```
r = session.get('web address')
```

r contains all the data sent from the site's server, including the HTML of the website

◎ Parse HTML and Tag Search

```
tag = r.html.find(selectors, first=True)
```

selectors are written in the manner of CSS Selector (for example, '#about' means to find the tag with the ID about), how to define the search criteria: using the inspect function of the Web browser

first=True this means returning only the first tag found, *first=False* returns the list containing all the found tags

From the found tag, it is possible to call *.find(...)* to find next in this tag

◎ Retrieving tag elements

tag.html: tag's HTML string

tag.text: tag's text string

tag.attrs: dictionaries containing tag attributes

Collecting data using Web APIs

- ◎ Some websites offer API (Application Programming Interface) to make external apps retrieve data easier
- ◎ Use the web API "more official" than parse HTML
 - As this is the path that "host" opens to "guests" entering the data
→ If the site has API, use it first.
- ◎ Need to read the host's document to know what data to take, which way to go, ...
- ◎ This is a list (incomplete) of sites providing API
 - <https://github.com/public-apis/public-apis>
 - Large sites like Twitter, Facebook, Google, ... Often provide API
 - Some sites require registration to use the API (charges may apply)

After collect data

- ◎ Whenever you collect data, you end up with **a group of values/scores on one or more variables.**
- ◎ What do you do next?
 - Understanding
 - Statistics
 - Visualization
 - Preprocessing
 - ...



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 - Visualization

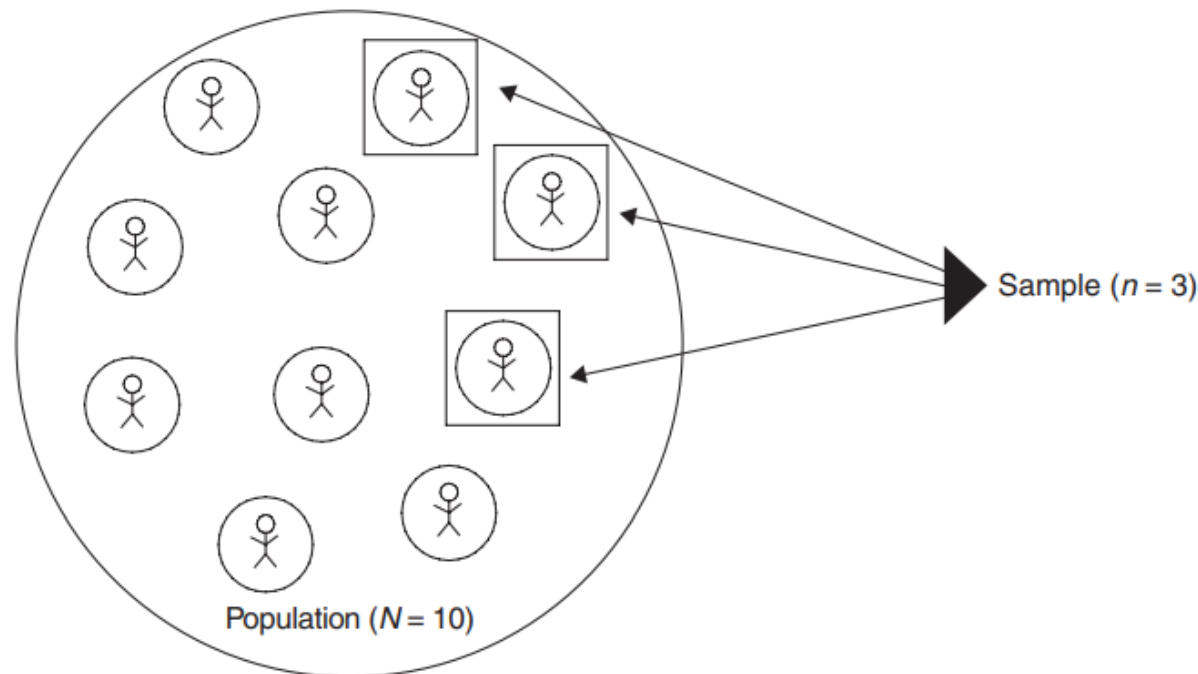
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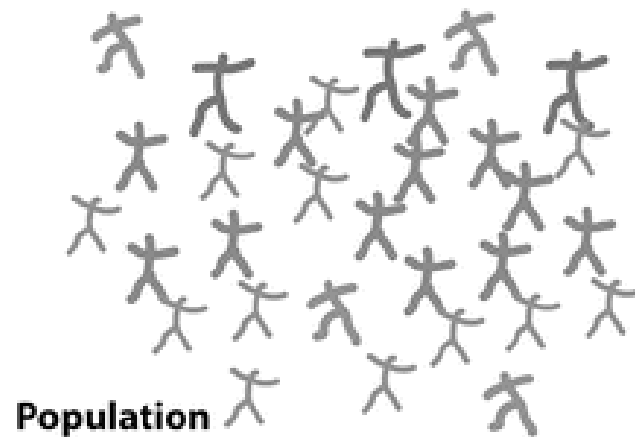
Population vs. Sample

- ◎ A **population** is an individual or group that represents **all the members** of a certain group or category of interest.
- ◎ A **sample** is a **subset drawn** from the larger population.
 - Randomly select a subset is known as a **random sample**.

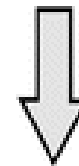
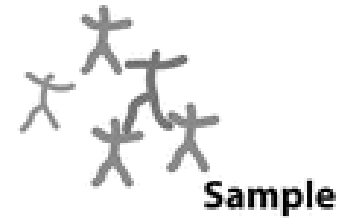


Population vs. Sample

- ◎ A population...
 - **Parameters** are numerical descriptions of **population** characteristics.
- ◎ A sample ...
 - **Statistics** are numerical descriptions from **sample data**.



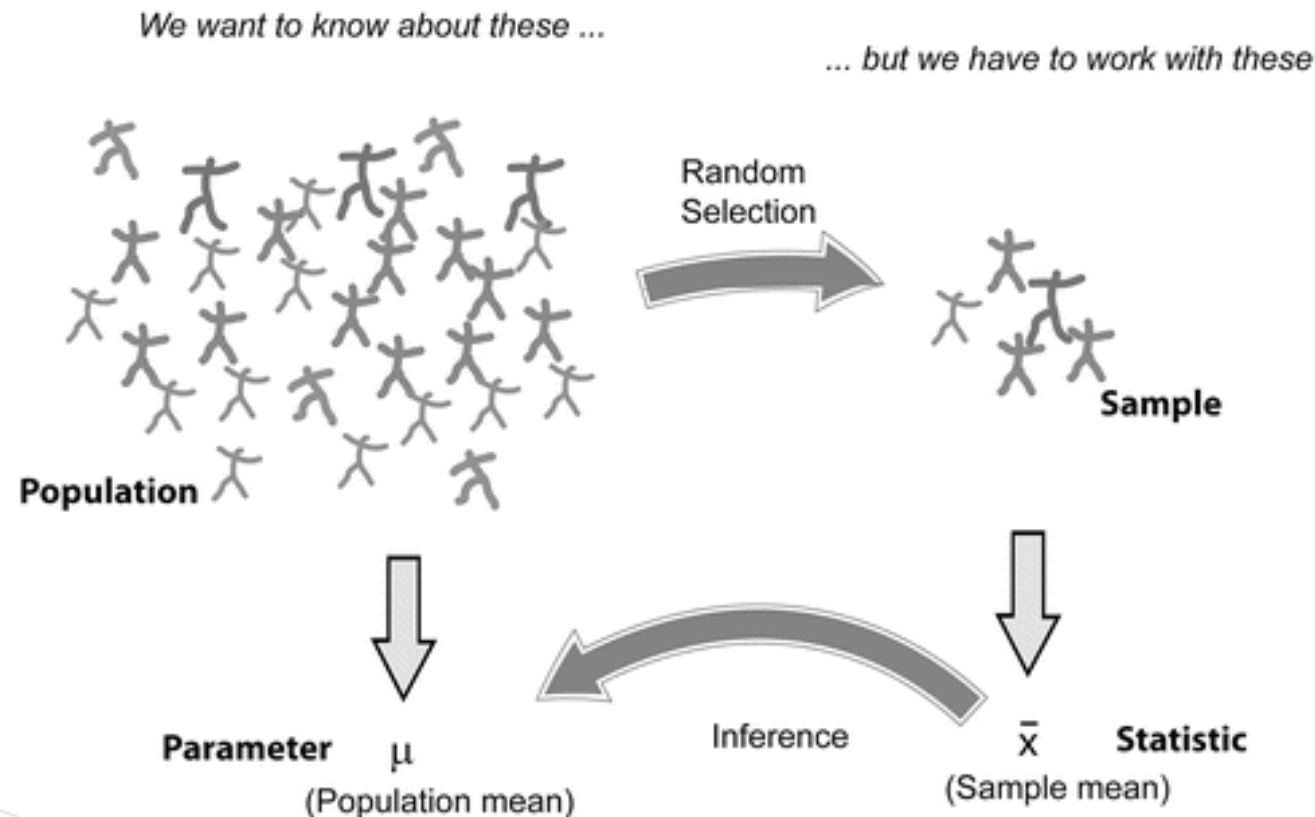
Parameter μ
(Population mean)



$\bar{x}$ **Statistic**
(Sample mean)

Descriptive vs Inferential Statistics

- © **Descriptive statistics** apply only to the members of a sample or population from which data have been collected.
- © **Inferential statistics** refer to the use of sample data to reach some conclusions about the characteristics of the larger population that the sample is supposed to represent.



What if ...?

© What if the sample is not truly representative of the population?

- We cannot be confident that conclusions based on our sample data will apply to the larger population.

→ **Sampling Issues**

Sampling

© Ways to select samples:

- **Random sampling**: the most useful, but also the most difficult
 - *Random* means that **every member** of a defined population has an **equal chance of being selected** into a sample.
 - Unbiased or biased sampling?



Sampling

◎ Ways to select samples:

- **Representative sampling:** selects cases so that they will match the larger population on specific characteristics.
- Unbiased or biased sampling?



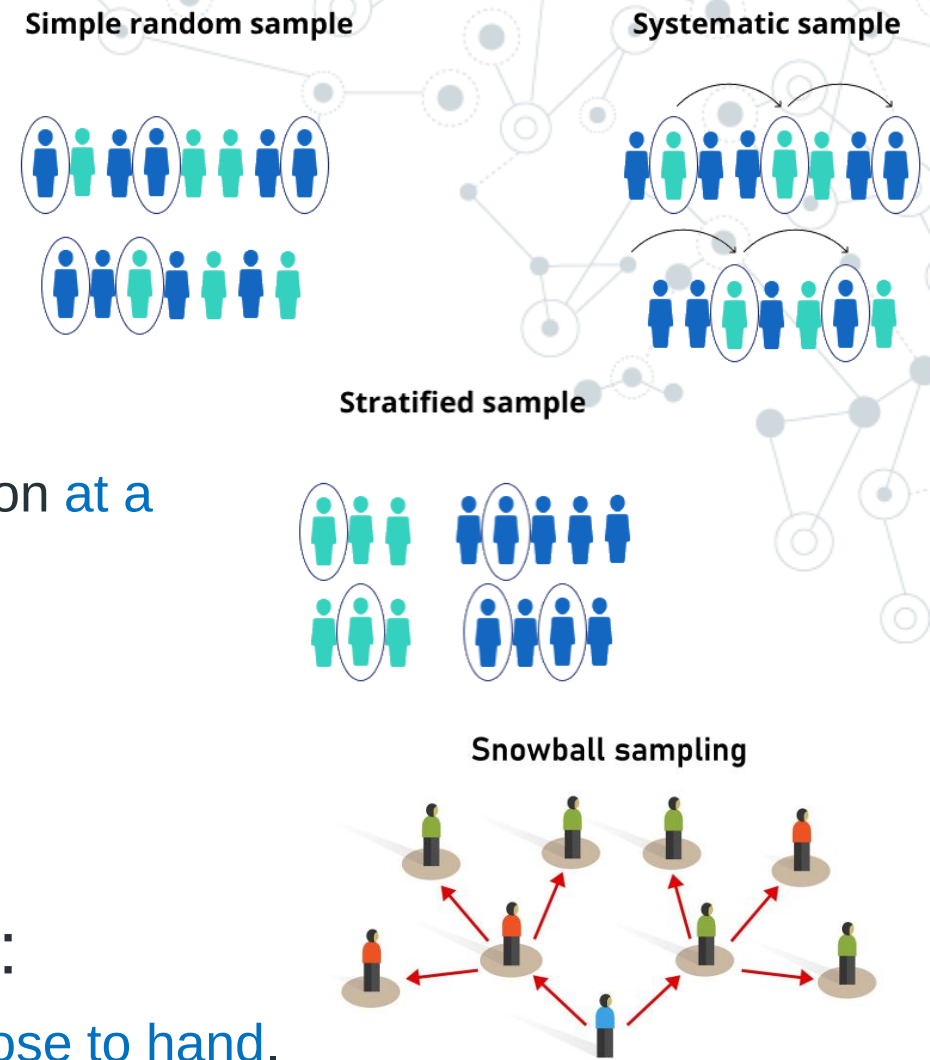
Other views about sampling

◎ Probability Sampling Techniques:

- Random sampling
- **Systematic sampling**: select members of the population at a regular interval.
- Stratified sampling: select specific proportions of individuals from various subpopulations (strata).

◎ Non-probability Sampling Techniques:

- Convenience sampling: select members which is close to hand.
- Snowball sampling: participants recruit other participants for a test.
- Quota sampling: decide quotas so that the samples can be useful in collecting data

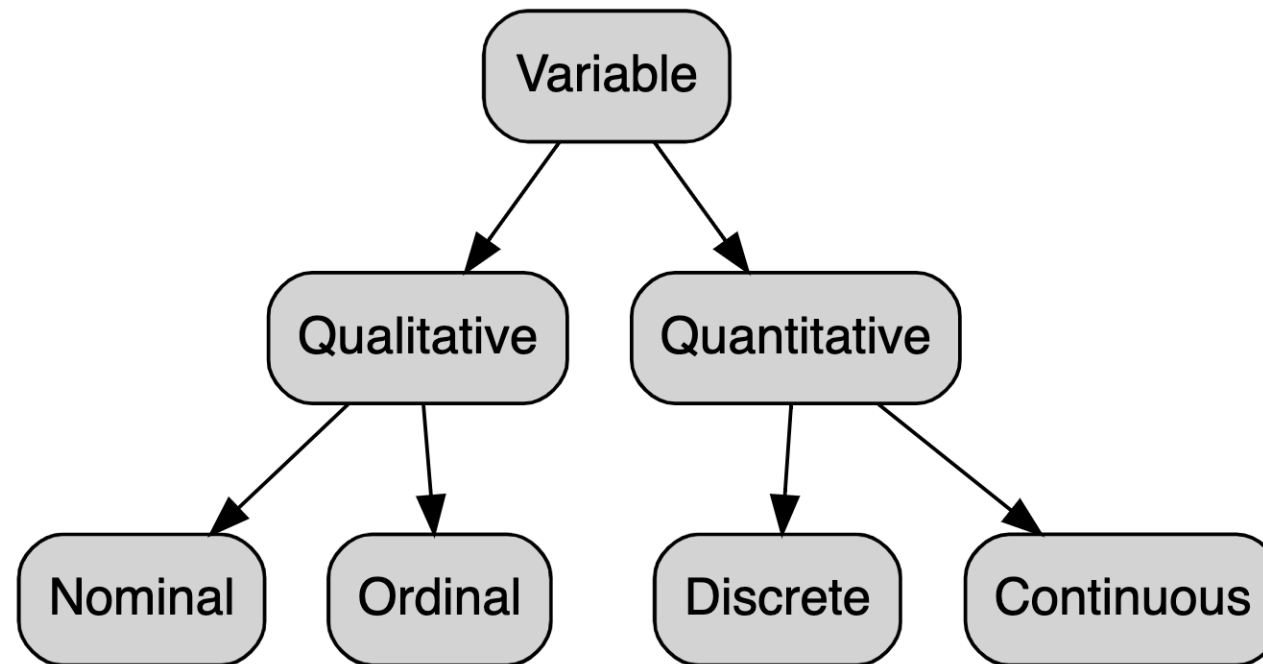


Variable and constant

- ◎ A **variable** is pretty much anything that can be codified and have more than a single value.
 - Example: income, gender, age, height, etc.
- ◎ A **constant** has only a single score.
 - Example: male, female, 25, etc.

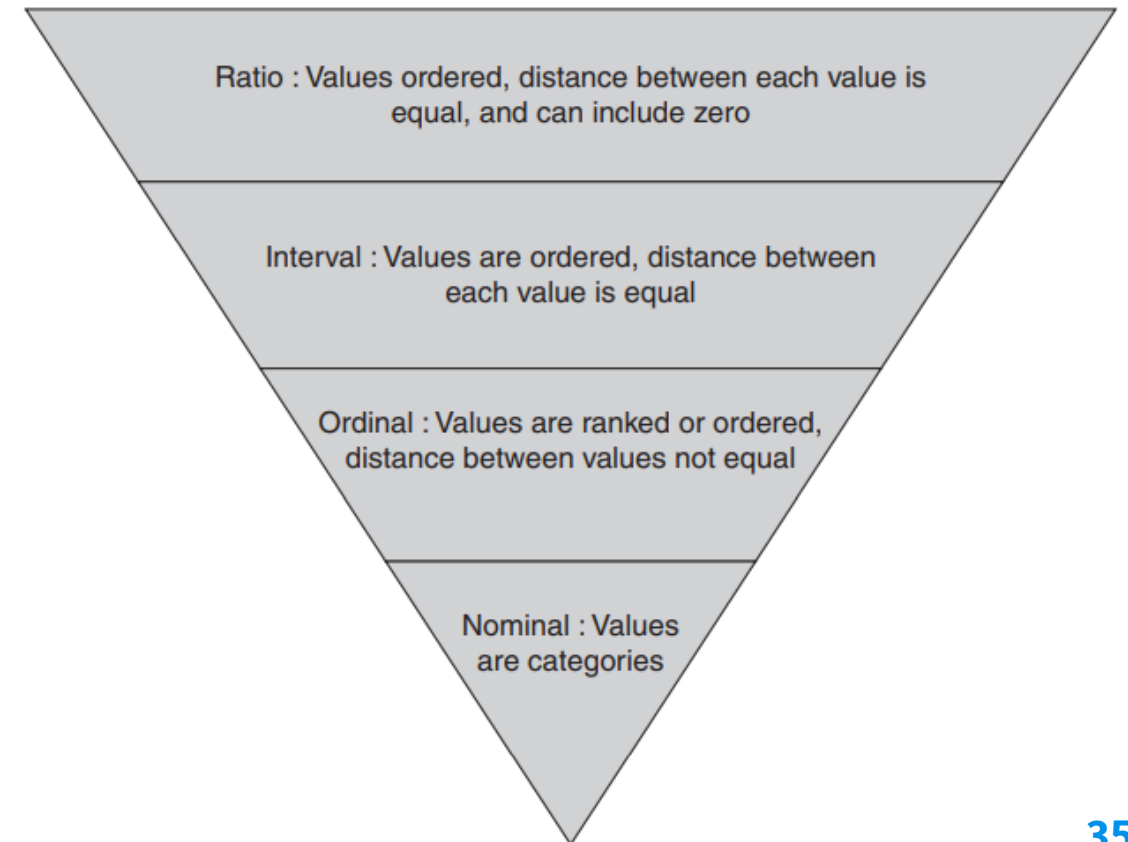
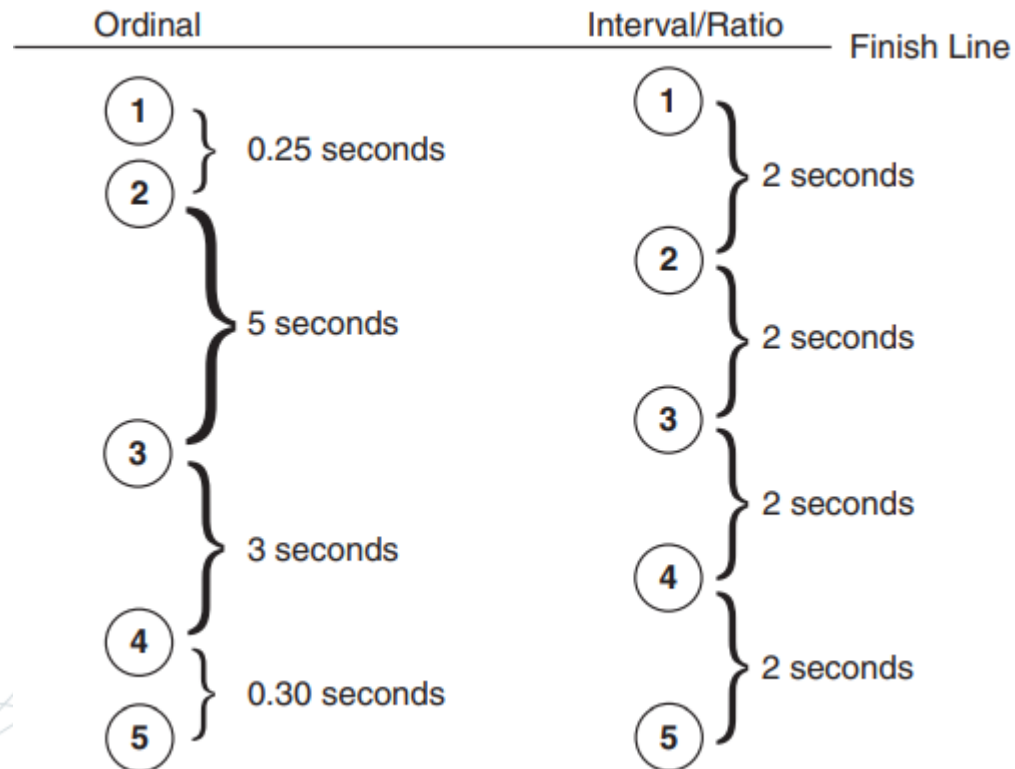
Types of variables

- © **Types of variables** include **qualitative** (nominal, ordinal categorical) and **quantitative** (discrete, continuous).



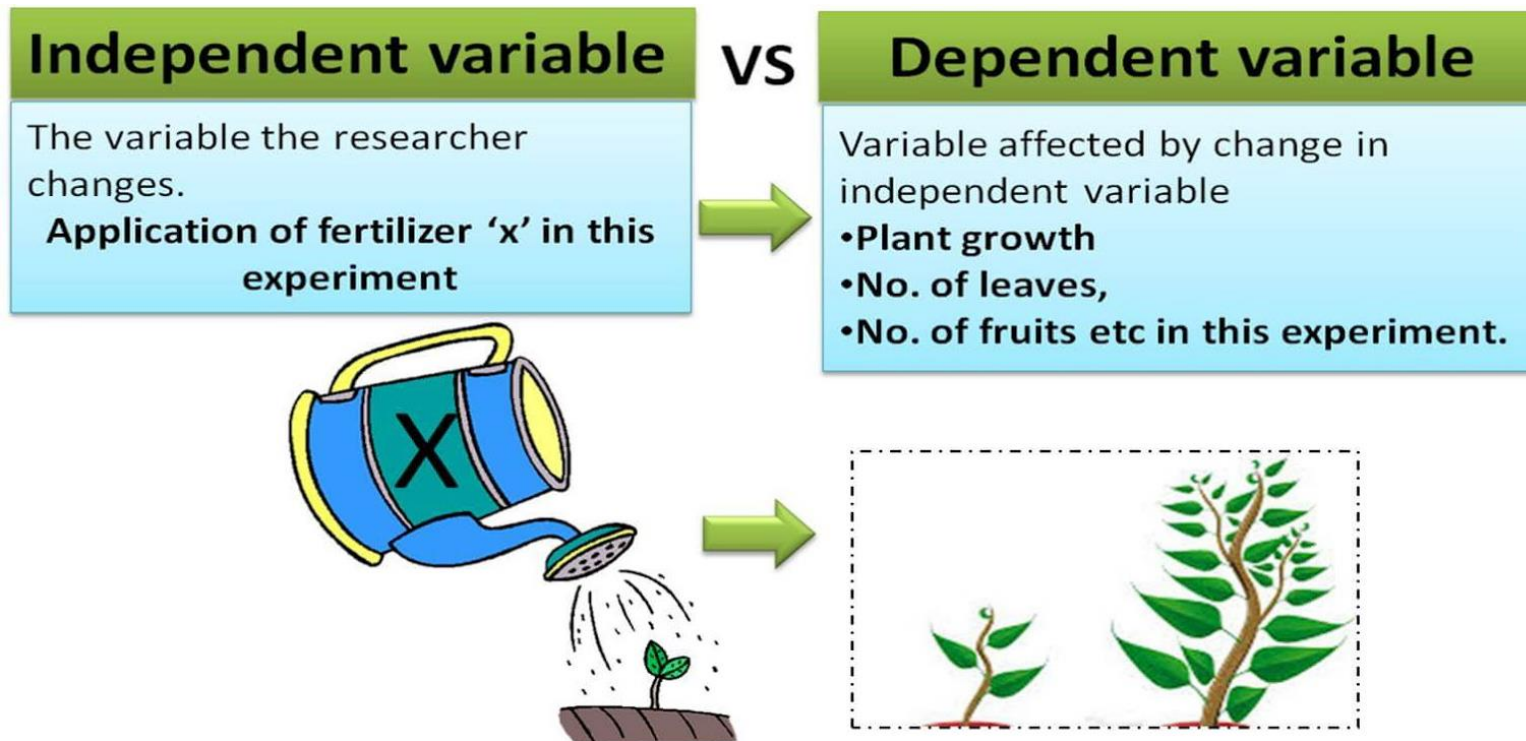
Scales of measurement

- Qualitative variables has 4 scales: nominal, ordinal, interval, and ratio.



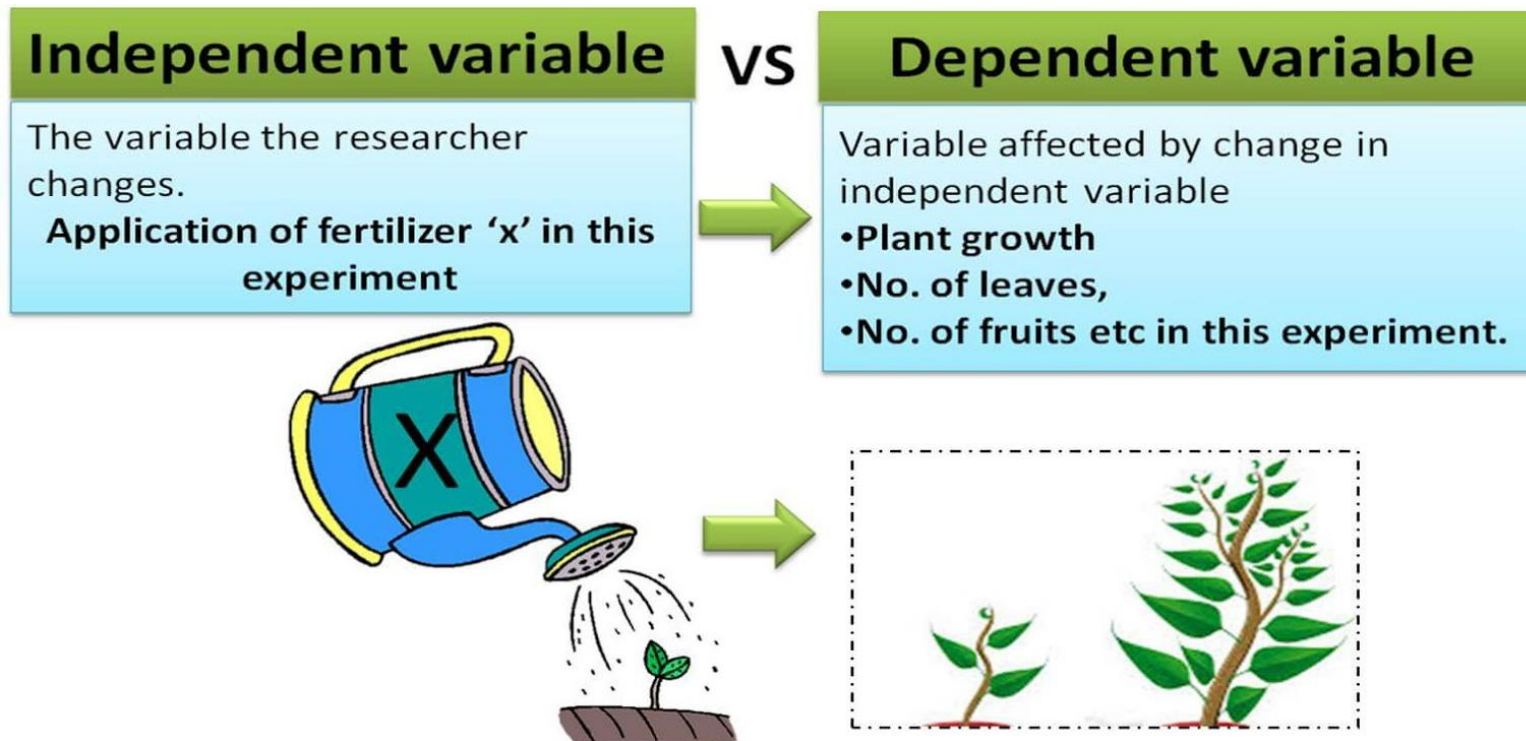
Independent vs dependent variables

- © The **independent variable** is the variable the experimenter **manipulates** or **changes**, and is assumed to have a **direct effect** on the **dependent variable**.



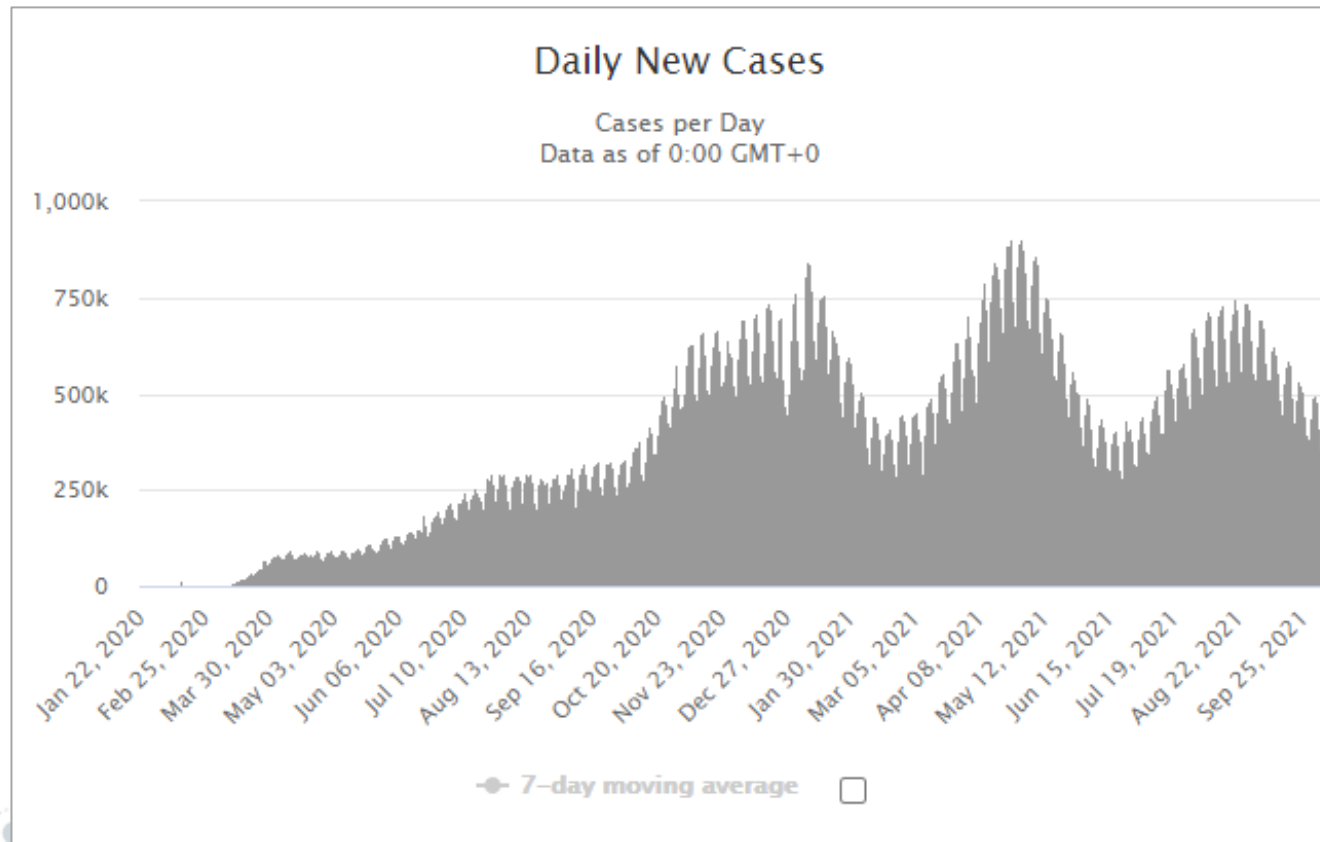
Independent vs dependent variables

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Distribution

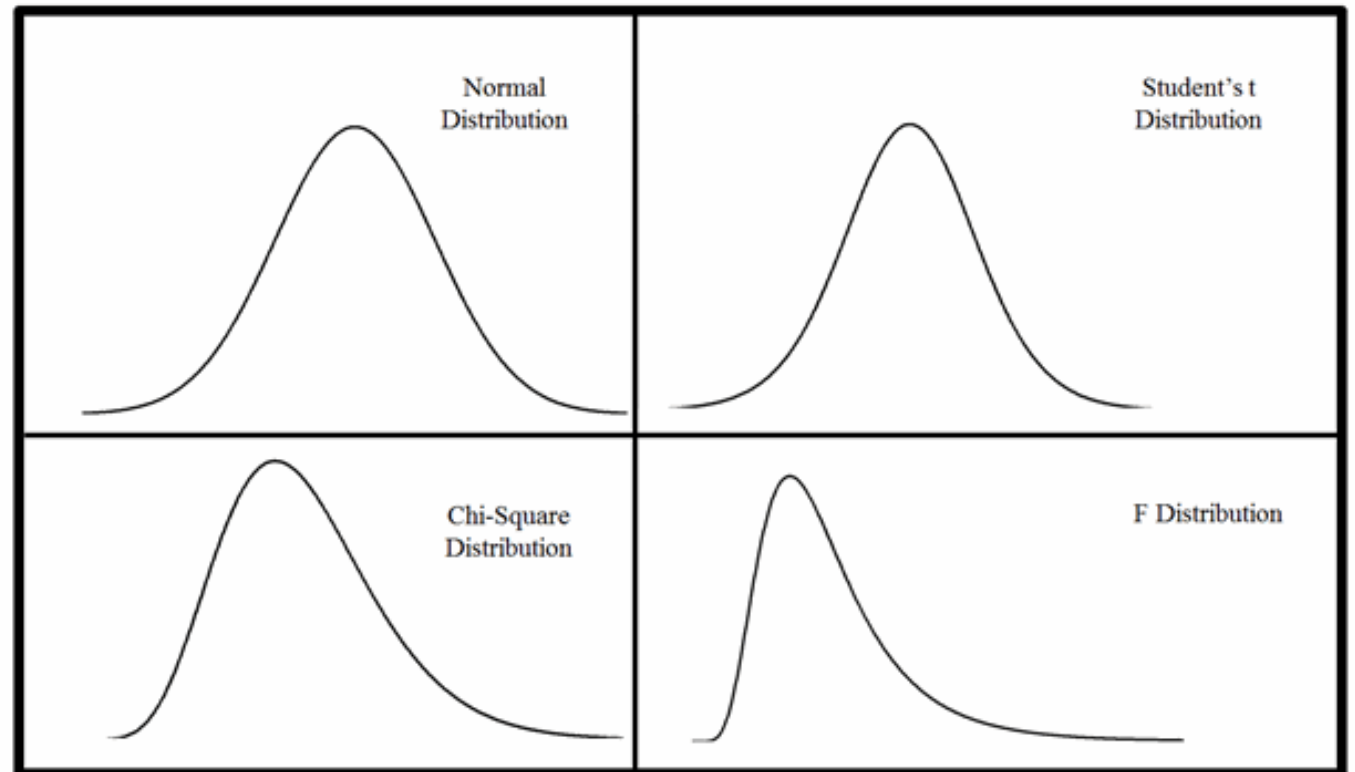
- © A **distribution** is simply a collection of data, or scores, on a variable which are arranged in order from smallest to largest and then they can be **presented graphically**.



Distribution Types

◎ There are many distribution types:

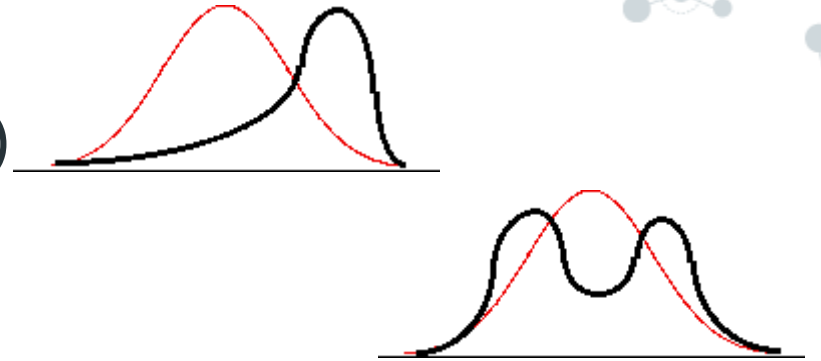
- Normal distribution
- t distribution
- F distribution
- chi-square distribution
- ...



Distribution characteristics

◎ Some distribution characteristics includes:

- Shape
- How spread out the value are (dispersion)
- max, min
- ...



◎ However, interested characteristics are **central tendency** and **variability**.

- Central tendency: **mean**, **median**, and **mode**.
- Variability: **range**, **variance**, and **standard deviation**.

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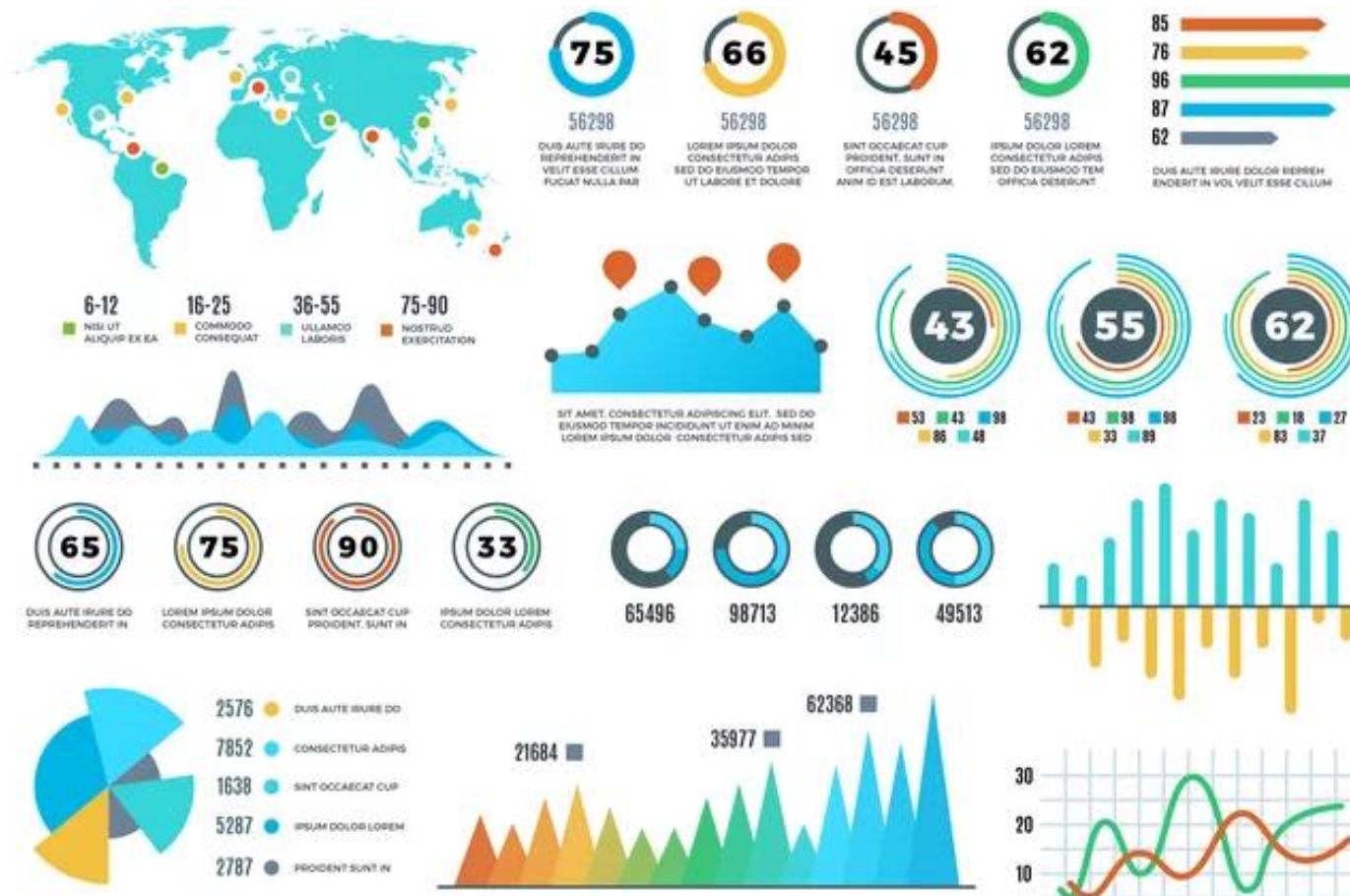
 - **Visualization**

- Preprocessing

- Machine learning

What is Data Visualization?

- ◎ **Data visualization** is the graphical representation of information and data.



Why is Data Visualization important?

- ◎ Visual data is **very easy to understand** compared to data in any other form.
 - Our brains process images at a rapid pace, according to an [MIT study](#).



Pros and Cons of Visualizing Data

◎ Pros:

- Complex data can be easily understood
- Simple sharing or communicating of data
- Easy to remember
- Storytelling can be done using dashboards and animations
- Data relations can be explored through interactive visualizations

◎ Cons:

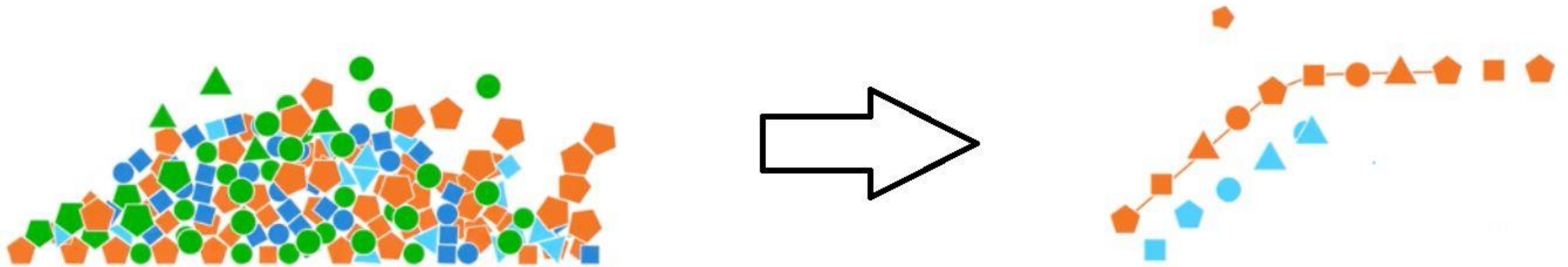
- It gives estimation not accuracy
- Biased
- Improper design issue
- Wrong focused people can skip core messages

Features of Visualizations

- ◎ All of data visualizations include the following features :
 - **Indicators**: They highlight the most important information.
 - **Simplicity**: The information is clear. The reader understands the information at hand immediately.
 - **Brevity**: The message is short and clear, and no unnecessary information is visible.
 - **Originality**: types of data are collected and displayed in a way that offers readers a new perspective on the subject.
 - **Colour**: to draw the reader's attention to the most important pieces of information, clear and easy-to-understand color palettes are used.

Data Wrangling

- ⦿ **Data wrangling**—also called **data munging**—is the process of transforming and mapping data from one "raw" data form into the format that is convenient for the consumption of data.



Tools and Libraries for Visualization

- ◎ Non-coding tool:
 - Tableau
 - Power BI
- ◎ Coding (Python, Matlab, R):
 - Python libraries:
 - ◎ Matplotlib
 - ◎ Seaborn
 - ◎ Geospatial
 - ◎ Bokeh

Types of Visualization

- ◎ **Comparison Plots:** include charts that are well-suited for comparing multiple variables or variables over time.
 - Line chart, bar chart, radar chart
- ◎ **Relation Plots:** are perfectly suited to show relationships among variables.
 - Scatter plot, bubble plot, correlogram, heatmap
- ◎ **Composition Plots:** are ideal if you think about something as a part of a whole.
 - Pie chart, stacked bar chart, venn diagram
- ◎ **Distribution Plots:** give a deep insight into how your data is distributed.
 - Histogram, density plot, box plot, violin plot
- ◎ **Geo Plots:** are a great way to visualize geospatial data.
 - Dot map, choropleth map, connection map

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Data quality

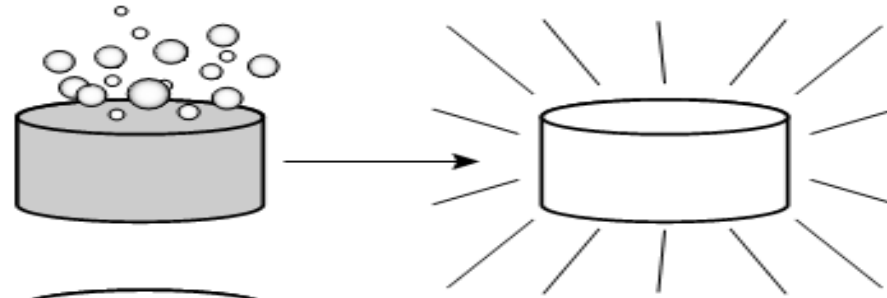
- ◎ **Missing, incomplete**: missing attribute value, missing attributes of interest, or only contains integrated data
 - Example : age, weight = " "
- ◎ **Noise**: contain errors or outliers
 - Example: salary = "-100 000"
- ◎ **Conflict**: there is inconsistency in the code or in the name
 - Example: age =42 , birth = 03/07/1997; US=USA?

Data quality

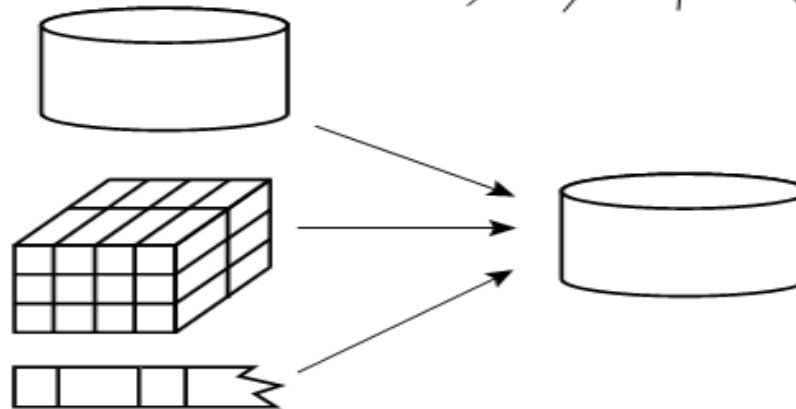
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Solutions? (1/2)

Data cleaning



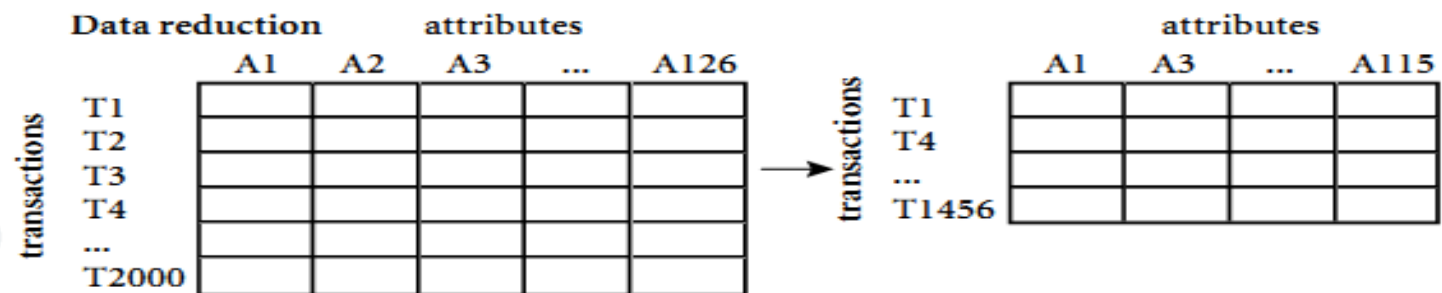
Data integration



Data transformation

$-2, 32, 100, 59, 48 \longrightarrow -0.02, 0.32, 1.00, 0.59, 0.48$

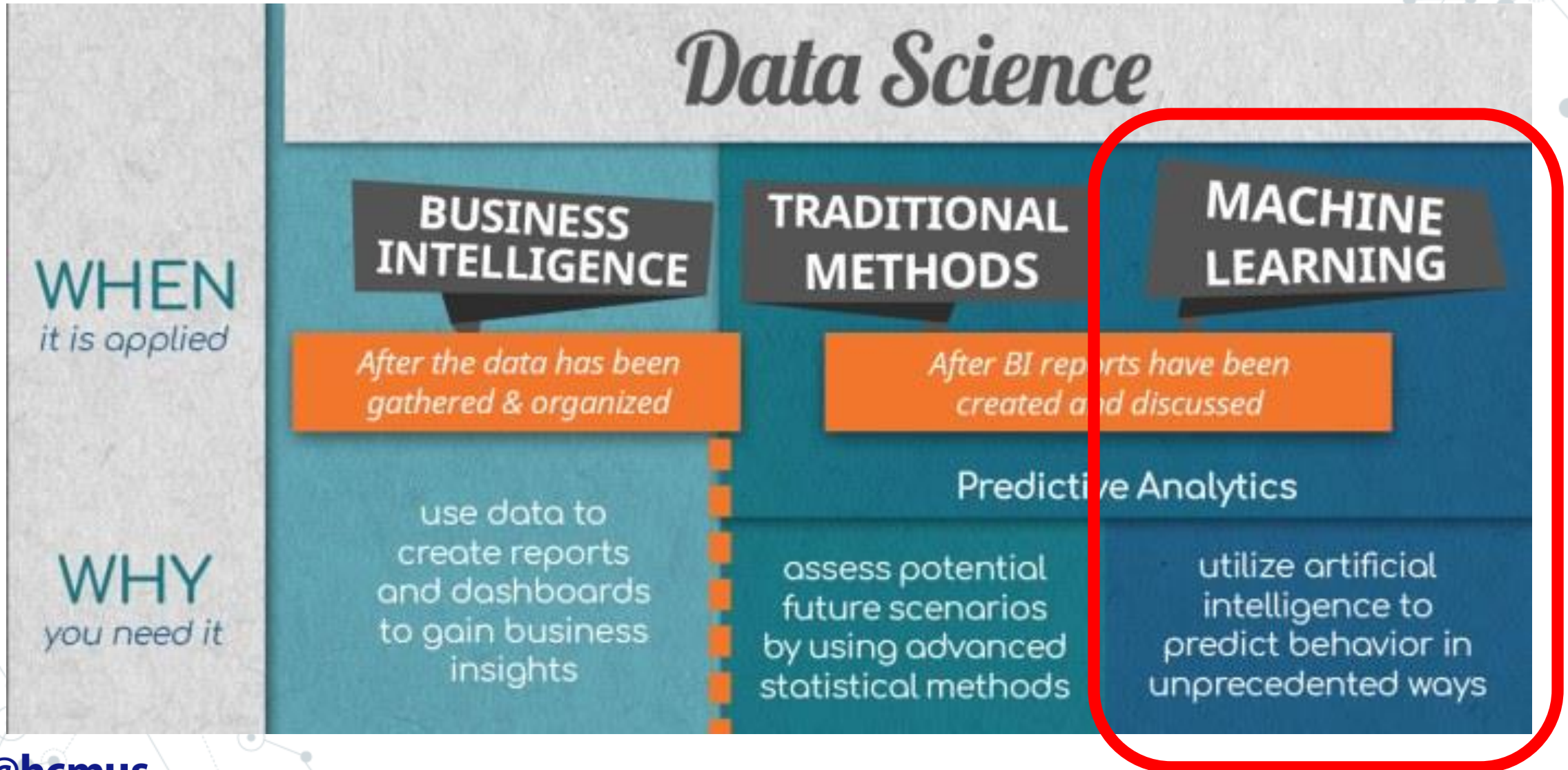
Data reduction



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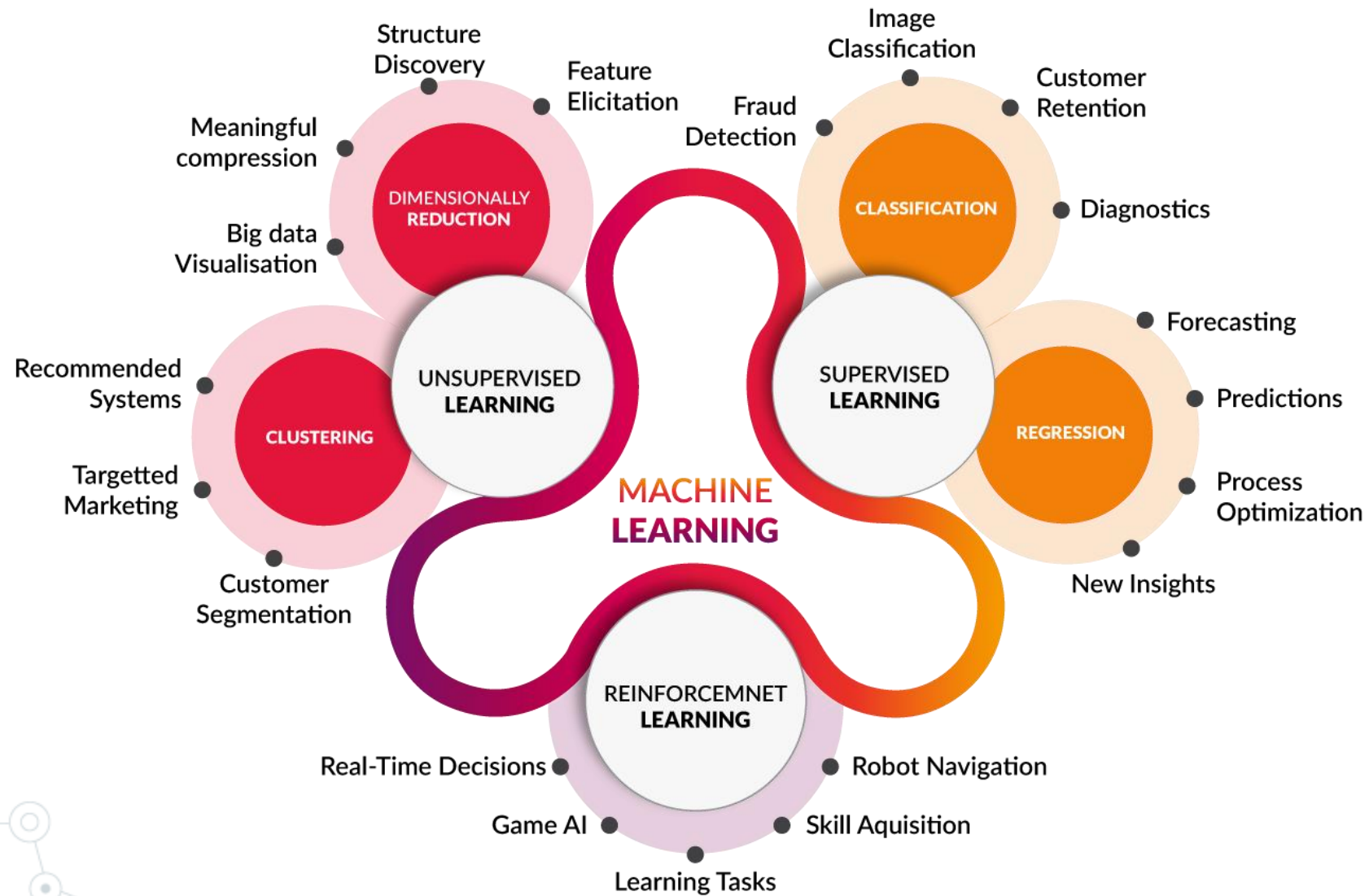
Data Science's tasks



Traditional Programming vs Machine Learning

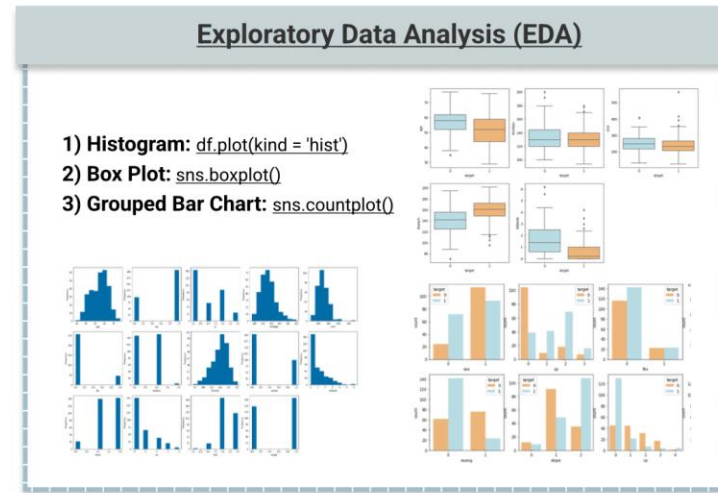


ML Tasks

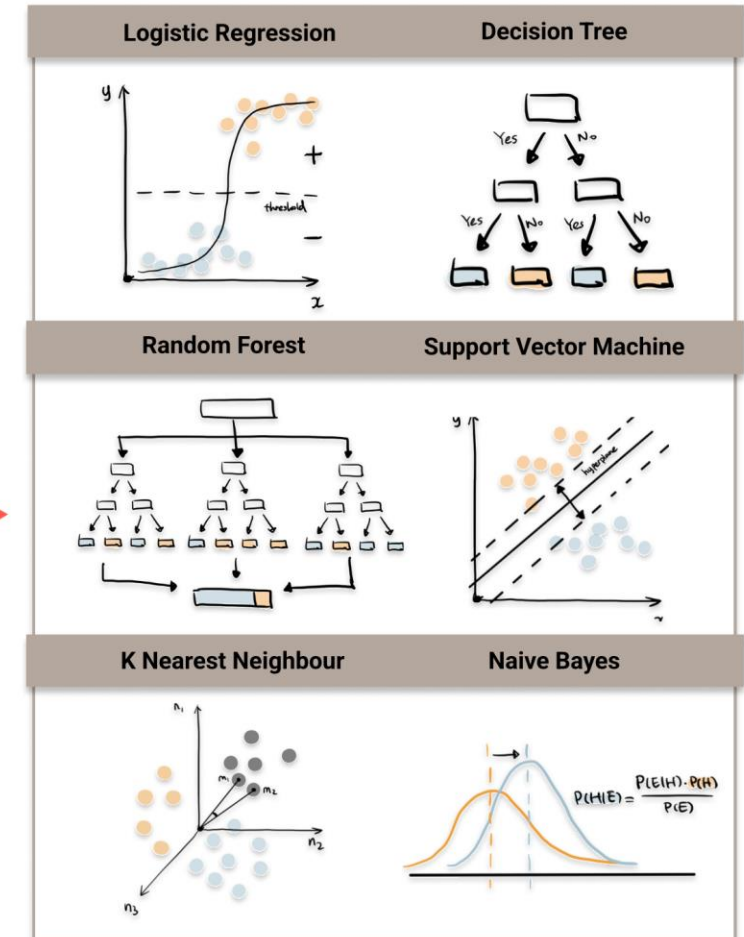


Machine Learning Choice

- Before implementing the machine learning (ML) model, the data scientist needs to **identify (several) branches** in ML that can solve the given problem.

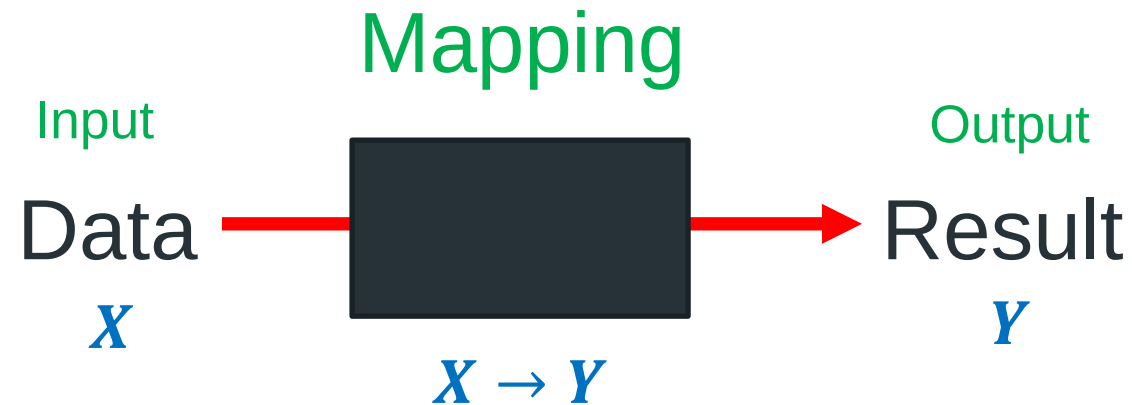


Visualization and
Statistics



Machine Learning

ML Architecture: Mapping



Learning = Identify mapping

How to identify the mapping

- ◎ Hypothesis
- ◎ Hypothesis
- ◎ Hypothesis
- ◎ ... and
- ◎ Hypothesis

You are here



Hypothesis Space

Hypothesis 1

Hypothesis 2

Hypothesis 3

Hypothesis 4

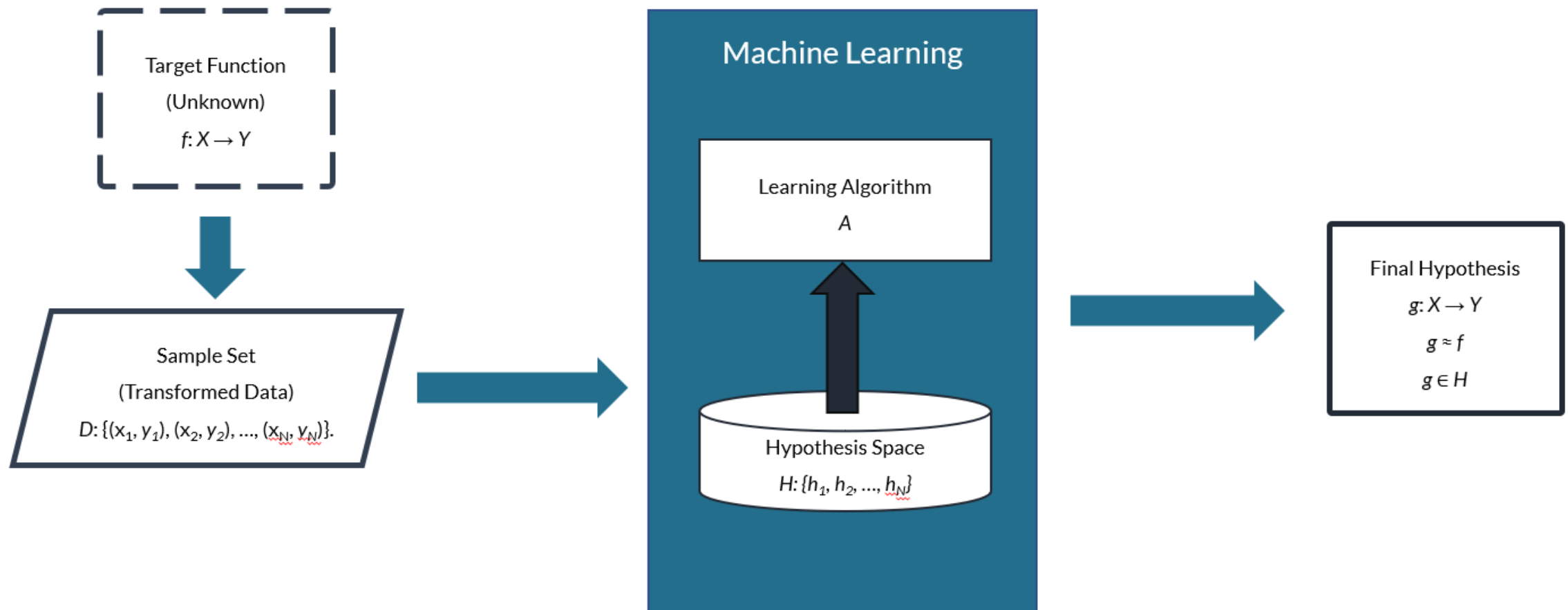
You want here



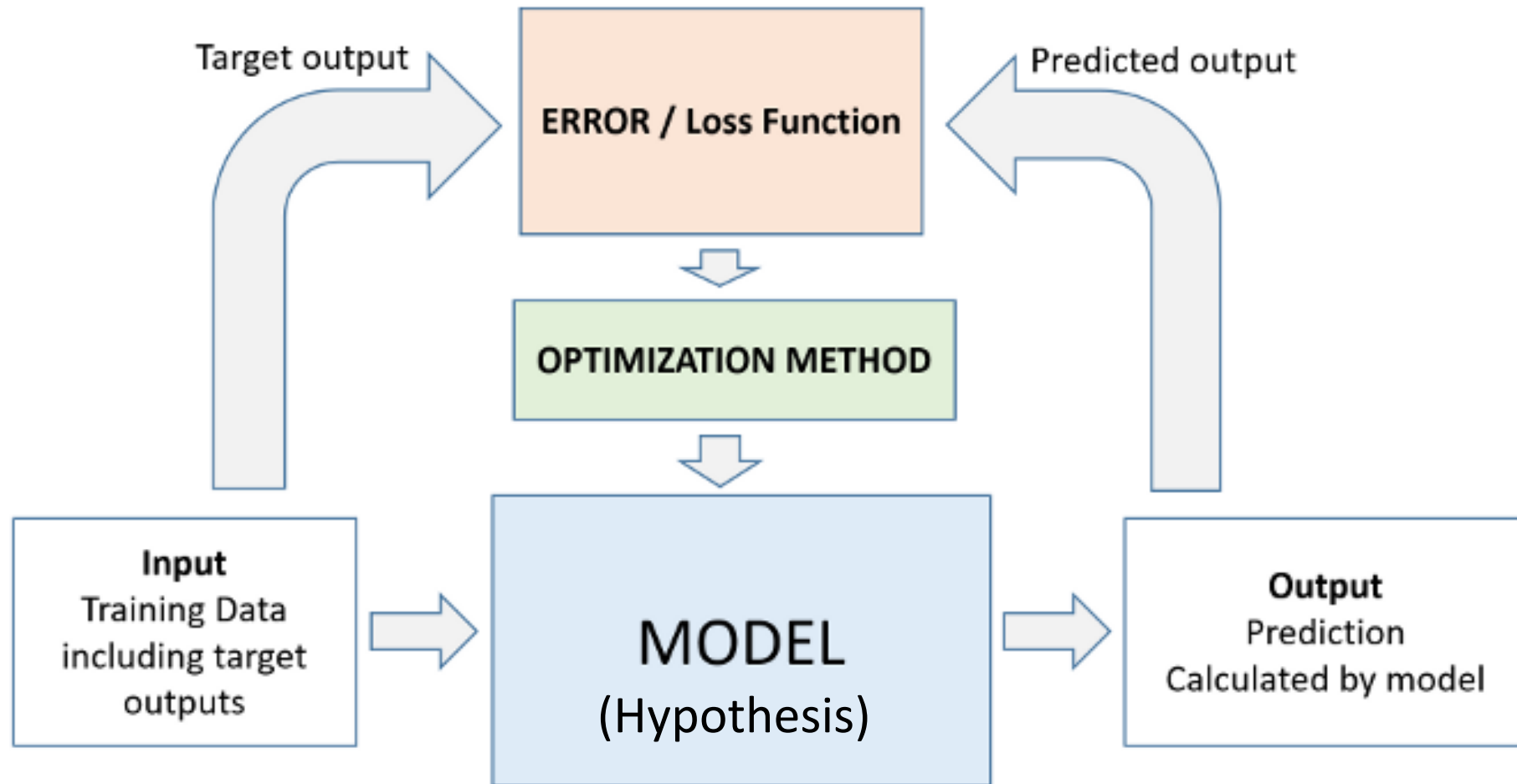
Best friend's hypothesis

Trying to Approximate

- © The purpose of ML is **trying to approximate a target function.**



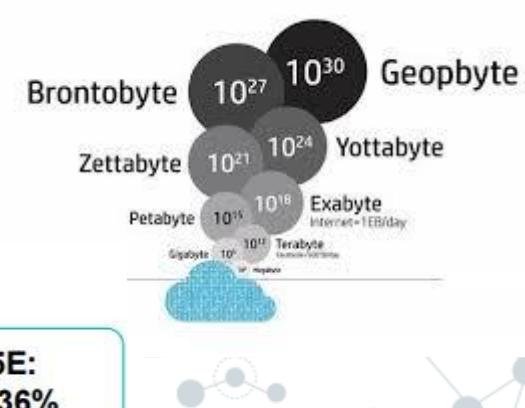
General model learning architecture



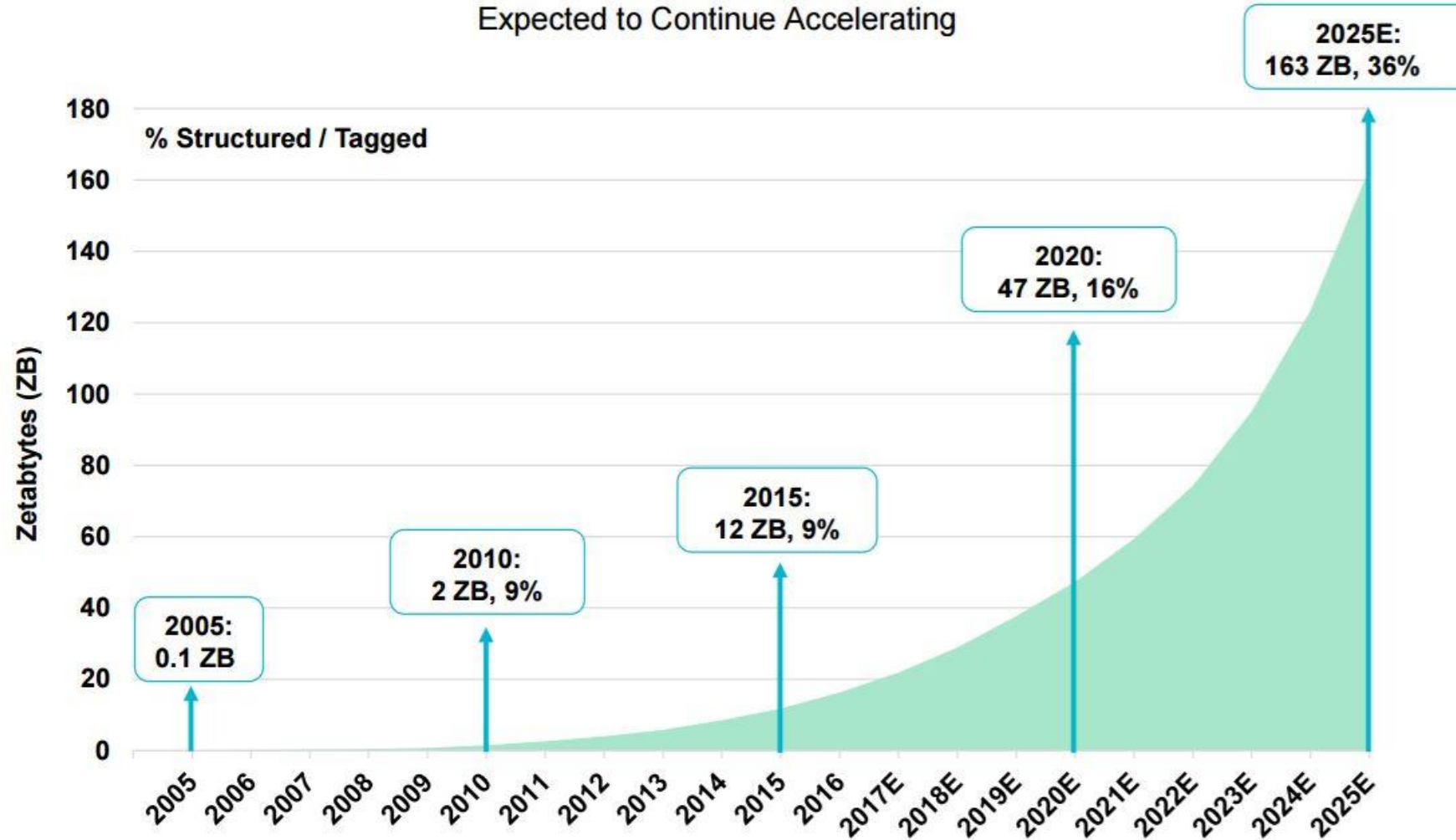
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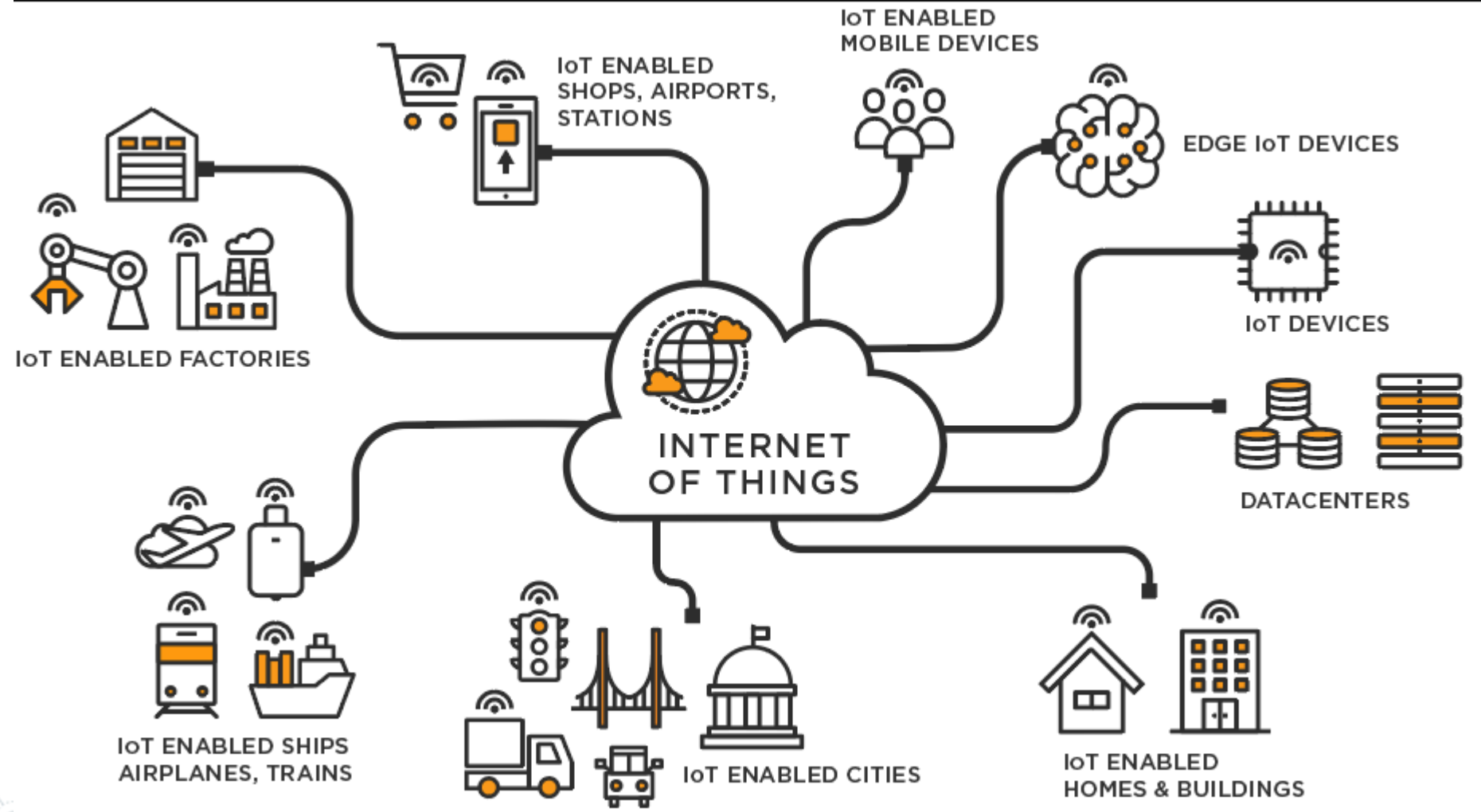
Data Growth



Information Created Worldwide =
Expected to Continue Accelerating



Data Never Sleeps



Number of hyper-parameters

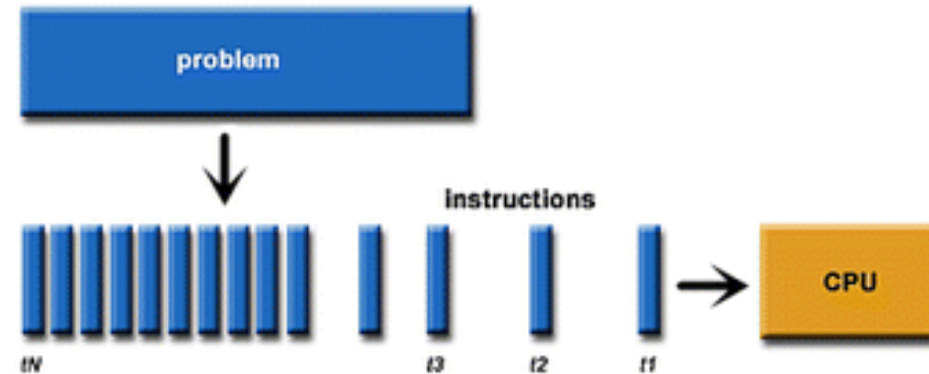
Model	Core differentiator	Pre-training objective	Parameters	Access	Information Extraction	Text Classification	Conversational AI	Summarization	Content generation
BERT	First transformer-based LLM	AE	370M	Source code					
RoBERTa	More robust training procedure	AE	354M	Source code					
GPT-3	Parameter size	AR	175B	API					
BART	Novel combination of pre-training objectives	AR and AE	147M	Source code					
GPT-2	Parameter size	AR	1.5B	Source code					
T5	Multi-task transfer learning	AR	11B	Source code					
LaMDA	Dialogue; safety and factual grounding	AR	137B	No access					
XLNet	Joint AE and AR	AE and AR	110M	Source code					
DistilBERT	Reduced model size via knowledge distillation	AE	82M	Source code					
ELECTRA	Computational efficiency	AE	335M	Source code					
PaLM	Training infrastructure	AR	540B	No access					
MT-NLG	Training infrastructure	AR and AE	530B	API					

Massive Data Analyzing Problem

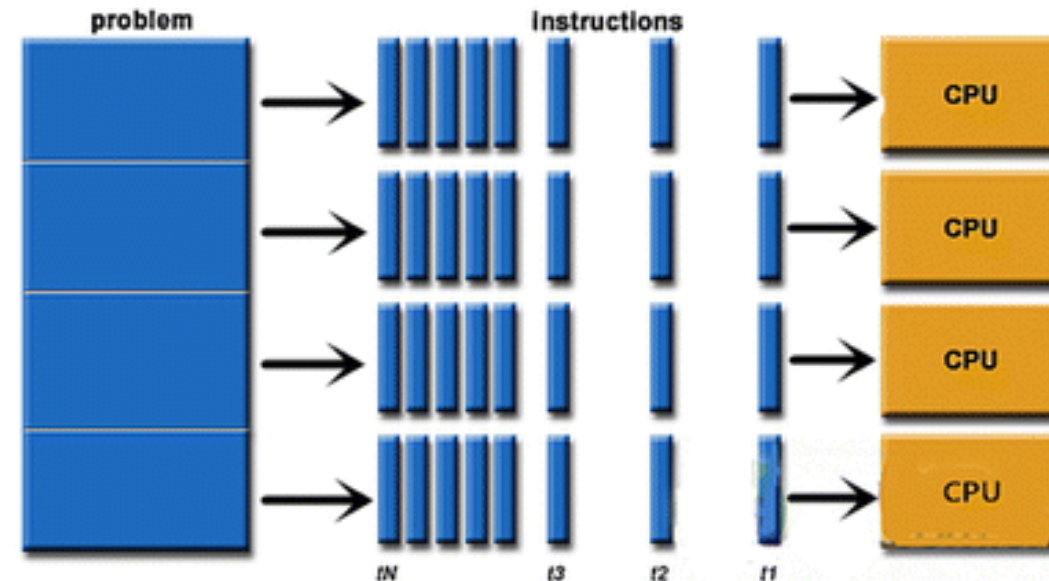


Parallel computing

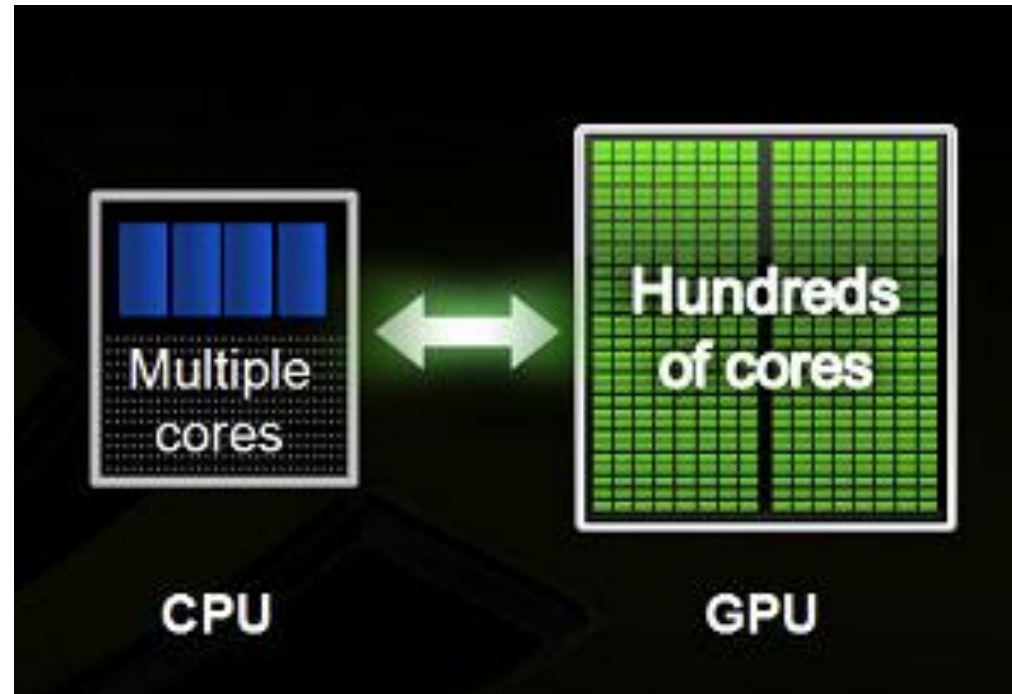
Serial operation schematic diagram



Parallel computing

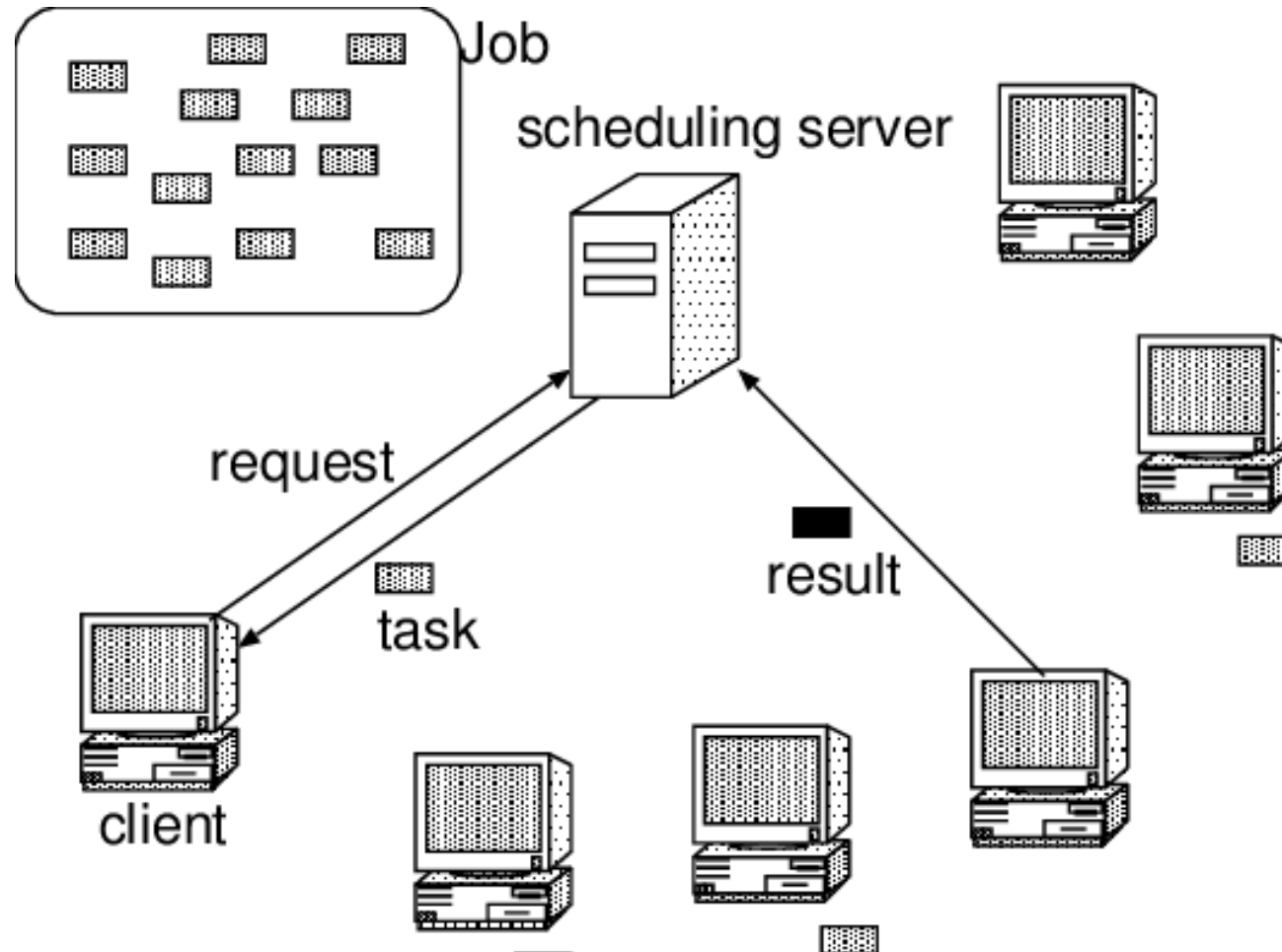


Parallel computing with GPU



To learn parallel programming,
enroll in the parallel programming
with GPU course

Distributed computing





The End